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Centro de Investigación en Ciencia Aplicada y Tecnología Avanzada
Unidad Querétaro

*Study of vegetable oil degradation using optical techniques
and artificial intelligence algorithms*

T H E S I S

Presented in order to obtain:

DOCTORADO EN TECNOLOGÍA AVANZADA

Presented by:

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Thesis advisors:

Dr. Jorge Adalberto Huerta Ruelas

Dr. Gonzalo Velazquez de la Cruz



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SECRETARÍA DE INVESTIGACIÓN Y POSGRADO

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1.- Se designa al aspirante el tema de tesis titulado:

Estudio de degradación de aceites mediante técnicas ópticas y algoritmos de inteligencia artificial

Objetivo general del trabajo de tesis:

Caracterizar la degradación de aceites fuera de línea mediante FT-IR y espectroscopia óptica (absorción y actividad óptica).
Desarrollar un sistema para monitoreo instantáneo de la degradación in situ de aceites
Desarrollar algoritmos de inteligencia artificial para cuantificar la degradación en espectros de FT-IR en aceites.

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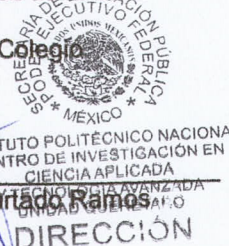
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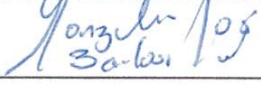
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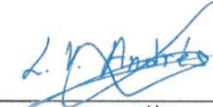
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Khemisset Marcos Escobar

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ABSTRACT

Thermal degradation of vegetable oils is a critical issue in both the food industry and domestic applications, as it directly affects nutritional quality, sensory properties, and product safety. During processes such as frying or prolonged heating, vegetable oils undergo chemical transformations driven by oxidation, hydrolysis, and the formation of secondary compounds, making reliable monitoring strategies essential. While spectroscopic techniques such as Fourier-transform infrared (FTIR) spectroscopy and ultraviolet–visible (UV–Vis) spectrophotometry provide rapid and non-destructive alternatives to conventional chemical assays, their performance strongly depends on appropriate spectral variable selection. In this thesis, we propose and validate a slope-based framework for objective feature extraction from FTIR spectra applied to the analysis of thermal degradation in vegetable oils. The method quantifies systematic absorbance changes at each wavenumber throughout the degradation process using linear regression, enabling identification of the most dynamically dynamic wavenumbers. Based on this criterion, a compact set of five representative wavenumbers is selected, providing an efficient description of the chemical evolution of the system while significantly reducing spectral dimensionality. The proposed framework is applied to the thermal degradation of soybean oil using ATR–FTIR and UV–Vis spectroscopy, supported by reference chemical measurements including iodine value (IV) and the specific extinction coefficients. The slope-selected wavenumbers are employed for unsupervised multivariate analysis, including principal component analysis and k-means clustering, allowing visualization of degradation trajectories and identification of degradation stages. Machine learning models are also developed to predict iodine value directly from the selected spectral variables, demonstrating that loss of unsaturation can be accurately estimated using a reduced feature set rather than full spectral information. The results indicate that the slope-based framework enhances interpretability, reproducibility, and computational efficiency compared to conventional band-based or region-based strategies. By preserving chemically meaningful information in a compact and data-driven manner, our framework provides a robust tool for spectroscopic monitoring of vegetable oil degradation. This work contributes to the development of more efficient analytical strategies for quality control and establishes a foundation for future spectroscopic studies of chemically evolving systems.

RESUMEN

La degradación térmica de los aceites vegetales es un problema crítico en la industria alimentaria y en aplicaciones domésticas, ya que afecta la calidad nutricional, las propiedades sensoriales y la seguridad del producto. Durante procesos como la fritura o el calentamiento prolongado, los aceites experimentan transformaciones químicas por oxidación, hidrólisis y formación de compuestos secundarios, lo que hace indispensable contar con estrategias de monitoreo confiables. Las técnicas espectroscópicas como FTIR y UV-Vis ofrecen alternativas rápidas y no destructivas, aunque su desempeño depende de una adecuada selección de variables espectrales.

En esta tesis se propone y valida un marco basado en pendientes para la extracción objetiva de características a partir de espectros FTIR aplicado a la degradación térmica de aceites vegetales. El método cuantifica los cambios de absorbancia en cada número de onda mediante regresión lineal, seleccionando un conjunto compacto de cinco números de onda representativos que describe eficientemente la evolución química del sistema y reduce la dimensionalidad espectral.

El marco se aplica a la degradación térmica de aceite de soya mediante ATR-FTIR y UV-Vis, respaldado por mediciones de referencia como el índice de yodo y los coeficientes de extinción específicos. Los números de onda seleccionados se emplean en análisis multivariante no supervisado —componentes principales y agrupamiento k-means— para visualizar trayectorias de degradación e identificar sus etapas. Además, se desarrollan modelos de aprendizaje automático para predecir el índice de yodo a partir de las variables espectrales seleccionadas.

Los resultados muestran que el marco basado en pendientes mejora la interpretabilidad, reproducibilidad y eficiencia computacional frente a estrategias convencionales. Al preservar información químicamente significativa de forma compacta, constituye una herramienta robusta para el monitoreo espectroscópico de la degradación de aceites vegetales y sienta las bases para futuros estudios de sistemas en evolución química.

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Chapter 1

INTRODUCTION

1.1 Background and motivation

Edible vegetable oils are essential components of human activities and play a central role in the food industry due to their nutritional value, sensory properties, and wide technological applications. Their composition is dominated by triglycerides containing saturated, monounsaturated, and polyunsaturated fatty acids, whose distribution determines both nutritional quality and oxidative stability. Among them, soybean oil and olive oil are of relevance because of their global consumption and economic importance.

Vegetable oils are chemically unstable under thermal, oxidative, and photochemical conditions. Oxidation and hydrolysis lead to the degradation of unsaturated fatty acids and to the formation of peroxides, aldehydes, and other secondary products that compromise oil quality and safety [1, 2]. In industrial and domestic processes involving prolonged heating or exposure to light and oxygen, reliable monitoring of these progression-based chemical changes is therefore critical.

Spectroscopic techniques, particularly Fourier-transform infrared (FTIR) spectroscopy and ultraviolet–visible (UV–Vis) spectrophotometry, have been widely applied to investigate oil degradation due to their speed, non-destructive nature, and sensitivity to molecular changes [3–5]. Conventional analyses often rely on predefined spectral markers, such as the carbonyl stretching band near 1745 cm^{-1} or absorbance values at 232 and 270 nm associated with conjugated dienes and trienes [6, 7]. While effective in specific contexts, these approaches are often system-dependent and show limited reproducibility when applied across different oils, experimental conditions, or materials [8]. These limitations arise in part because the progression systems as the carbonyl stretching band near 1745 cm^{-1} or absorbance values at 232 and 270 nm associated with conjugated dienes and trienes. While effective in specific contexts, these approaches are often system-dependent and show limited reproducibility when applied across different materials, experimental conditions.

Recent advances in data science and machine learning offer new opportunities for extracting reproducible information from complex spectroscopic datasets [9, 10]. However, many existing studies still depend on empirical variable selection or broad spectral regions, which may overlook chemically informative features and limit model generalizability [11, 12]. This points to the need for data-driven frameworks capable of objectively identifying dynamic wavenumbers while maintaining chemical interpretability across progression systems.

Throughout this thesis, the term progression system refers to any spectroscopic dataset acquired along a continuous variable, such as time, temperature, processing cycles, or mechanical exposure, in which chemical or physical changes evolve in a directional and measurable manner. This definition encompasses the thermal degradation of edible oils as the primary application but extends to any system where spectral evolution can be tracked quantitatively.

1.2 Problem Statement

The accurate monitoring of oil degradation remains a challenge in both research and industry. Traditional chemical assays such as iodine value determination, peroxide value, or p-anisidine index provide reliable information on the oxidative state of oils, but they are labor-intensive, destructive, and unsuitable for continuous monitoring. Although spectroscopic methods such as FTIR and UV-Vis offer rapid and non-destructive alternatives, their effectiveness depends largely on the correct identification of spectral variables that reflect chemical transformations.

Current approaches typically rely on predefined bands or integrated spectral regions that are selected based on literature reports or empirical experience. While these strategies can identify well-known markers such as the carbonyl group near 1745 cm^{-1} or conjugated diene absorption at 232 nm, they lack generalizability when applied to different oils, experimental conditions, or degradation mechanisms. For example, Han et al. [8] demonstrated that fingerprint regions effective for classifying Camellia oil blends required manual re-selection when extended to other oil types, illustrating how fixed-band strategies constrain transferability. This dependence on fixed spectral regions therefore introduces subjectivity, limits reproducibility, and reduces the transferability of results across systems.

Furthermore, the high dimensionality of FTIR spectra, combined with the collinearity between adjacent wavenumbers, complicates the extraction of meaningful features for multivariate analysis. Chemometric methods such as principal component analysis and partial least squares regression have been widely adopted to address this issue, yet they still rely on the quality of the initial variable selection. Without a systematic and data-driven strategy, there is a risk of overlooking subtle but chemically relevant features that could improve interpretability and predictive performance.

The lack of robust and reproducible feature selection also hinders the integration of spectroscopic data with machine learning models aimed at predicting chemical reference values such as the iodine value. Studies such as Jamwal et al. [3] and Nascimento et al. [13] achieved accurate predictions within their calibration sets, but model performance degraded when applied to oil types or spectral conditions outside the original training domain. As a result, the development of predictive models has been limited by noisy inputs, overfitting, and reduced accuracy when applied to external datasets. Addressing this limitation requires a methodological framework capable of identifying

dynamic and chemically meaningful spectral variables while maintaining generalizability across oils and experimental conditions.

1.3 Hypotheses

Our central hypothesis is that a slope-based framework for spectral variable selection can objectively identify the most dynamic and informative wavenumbers in progression system FTIR datasets by quantifying systematic absorbance changes through regression analysis.

A second hypothesis is that slope-selected FTIR variables, when used as inputs for machine learning regression models, enable accurate prediction of chemical reference parameters, such as iodine value in oils, providing a compact, interpretable, and generalizable alternative to full-spectrum approaches across different progression systems.

1.4 Research Objectives

Our general objective is to develop and validate a slope-based framework for the analysis of FTIR and UV-Vis spectroscopic data to monitor and predict the degradation of vegetable oils. The framework is designed to provide a reproducible, data-driven strategy for variable selection and to integrate spectroscopic information with machine learning models for the prediction of reference chemical indices.

To achieve this goal, the work is structured around several specific objectives. The first is to establish experimental protocols for controlling thermal degradation of soybean oil, generating time-resolved FTIR and UV-Vis datasets as well as reference measurements of iodine value and UV absorbance indices. The second is to implement a systematic preprocessing pipeline including smoothing, normalization, and derivative transformation to ensure consistent data quality. The third is to design and apply a slope-based framework for variable selection, quantifying the rate of spectral change as a means of identifying dynamic and chemically meaningful wavenumbers. The fourth is to validate the proposed framework by comparing it against conventional selection methods and by applying it to external datasets that show progression systems (polymers, lubricants, and paints). The fifth is to construct and evaluate machine learning models, including regression and clustering approaches, to classify degradation stages and to predict the iodine value directly from slope-selected FTIR variables.

Together, these objectives aim to consolidate a methodological framework that bridges spectroscopy, chemometrics, and data science, with direct relevance for vegetable oil quality monitoring.

1.5 Scope and Limitations

Certain limitations are inherent to the scope of our work. The experiments were restricted to a single degradation temperature and to a limited number of vegetable oils, which may constrain the generalization of the results to other thermal conditions or oil types. The slope-based framework relies on linear regression to quantify spectral responsiveness, and therefore it may underrepresent non-linear or transient behaviors in the spectra. The prediction of iodine value was also based on laboratory-prepared datasets with limited sample size, which may affect model robustness when applied to larger and more diverse populations. Despite these limitations, the framework was designed to be modular and adaptable, providing a foundation for future extensions and industrial applications.

Chapter 2

LITERATURE REVIEW

2.1 Chemical Mechanisms of Oil Degradation

The stability of edible oils is primarily determined by the composition of their fatty acids, with a higher degree of unsaturation associated with increased susceptibility to degradation. Polyunsaturated fatty acids such as linoleic (C18:2) and linolenic (C18:3) are particularly prone to oxidative reactions, while monounsaturated fatty acids such as oleic acid (C18:1) exhibit greater stability. Saturated fatty acids, including palmitic (C16:0) and stearic (C18:0), contribute to higher thermal resistance but reduce nutritional value when present in excess.

The degradation of oils occurs through three main pathways: oxidation, hydrolysis, and polymerization. Oxidation is the most critical process, leading to the formation of free radicals, peroxides, aldehydes, and other secondary oxidation products that alter both the chemical and sensory properties of oils. It is initiated by the abstraction of a hydrogen atom from a methylene group adjacent to a double bond, producing lipid radicals that react with molecular oxygen to form peroxy radicals. These species propagate a chain reaction that results in hydroperoxides, which subsequently decompose into aldehydes, ketones, and volatile compounds responsible for rancidity odor.

The oxidative stability of oils is also influenced by temperature, light, and the presence of pro-oxidant metals. Matthäus et al. [1] reported that high-temperature conditions significantly accelerate oxidation by promoting hydroperoxide breakdown and secondary reactions. Light, particularly in the ultraviolet range, induces photochemical reactions that degrade pigments such as chlorophyll and carotenoids, producing singlet oxygen that enhances the oxidation rate.

Hydrolysis, although less dominant under normal storage, occurs during frying and other high-temperature processes in the presence of water, leading to the release of free fatty acids and monoacylglycerols, which further destabilize the oil. Polymerization occurs mainly during repeated heating, forming high-molecular-weight compounds that increase viscosity and reduce nutritional value.

Several studies have examined the kinetics of oil degradation and the factors influencing stability. Frega et al. [2] demonstrated that free fatty acid content plays a critical role in oxidative resistance. Anwar et al. [14] observed significant differences in oxidative stability among soybean varieties due to variations in fatty acid profiles. More recently, Jamwal et al. [3] emphasized the role of FTIR spectroscopy in detecting chemical changes associated with oxidation, highlighting the relevance of

carbonyl stretching vibrations around 1740 cm^{-1} and C–H stretching near 2920 cm^{-1} as indicators of degradation. Similarly, Mousa et al. [4] provided an overview of FTIR applications for oil quality, authenticity, and adulteration detection, stressing the importance of integrating chemometric tools for robust interpretation.

Comprehensive reviews by Meenu et al. [5] and Cakmak-Arslan [6] confirmed that the degradation process is dynamic and system-specific, requiring analytical approaches capable of tracking subtle changes across time.

2.2 Spectroscopic Indicators in FTIR and UV-Vis

Spectroscopic methods provide a rapid and non-destructive means of monitoring the chemical transformations that occur during oil degradation. Among these, Fourier-transform infrared (FTIR) spectroscopy and ultraviolet–visible (UV-Vis) spectrophotometry have been extensively applied due to their sensitivity to structural changes in fatty acids and oxidation products.

In FTIR, characteristic vibrational bands serve as reliable indicators of degradation. The band around $1740\text{--}1745\text{ cm}^{-1}$, corresponding to the stretching of ester carbonyl groups, is widely used to monitor the formation of oxidation products and the breakdown of triglycerides. Similarly, the absorption near 3007 cm^{-1} is associated with =C–H stretching in cis-double bonds, and its reduction during heating reflects the loss of unsaturation in fatty acids. Bands around 722 cm^{-1} correspond to CH_2 rocking vibrations of long-chain hydrocarbons and are sensitive to changes in lipid packing. Other relevant regions include the C–O stretching vibrations near 1160 cm^{-1} and 1120 cm^{-1} , often linked to esters, and the CH stretching modes around 2924 and 2854 cm^{-1} , which reflect alkyl chain degradation.

The application of FTIR to oil quality assessment has been widely documented. Poiana et al. [7] demonstrated its ability to detect adulteration of olive oil with soybean oil and to monitor degradation at elevated temperatures. Han et al. [8] successfully applied FTIR with attenuated total reflectance (ATR) accessory combined with chemometrics to classify Camellia oil blends, highlighting the potential of fingerprint regions between 900 and 1200 cm^{-1} . More recently, Cakmak-Arslan [6] reported that monitoring hazelnut oil under thermal processing using FTIR allowed precise tracking of structural changes and classification of oils compared with olive oil. Mousa et al. [4] provided a comprehensive review of FTIR applications in detecting authenticity, oxidation, and adulteration of edible oils, concluding that chemometric integration significantly enhances reliability.

Figure 2.1 shows the specific FTIR spectrum of soybean oil, identifying the characteristic FTIR absorption peaks associated with key molecular vibrations that characterize the chemical composition of vegetable oils. These wavenumbers correspond to functional groups and structural components relevant to understanding the degradation and quality of the oil and are used to perform comple-

mentary spectral analysis, illustrating how FTIR wavenumbers serve as molecular fingerprints for quality assessment and thermal degradation monitoring in vegetable oils.

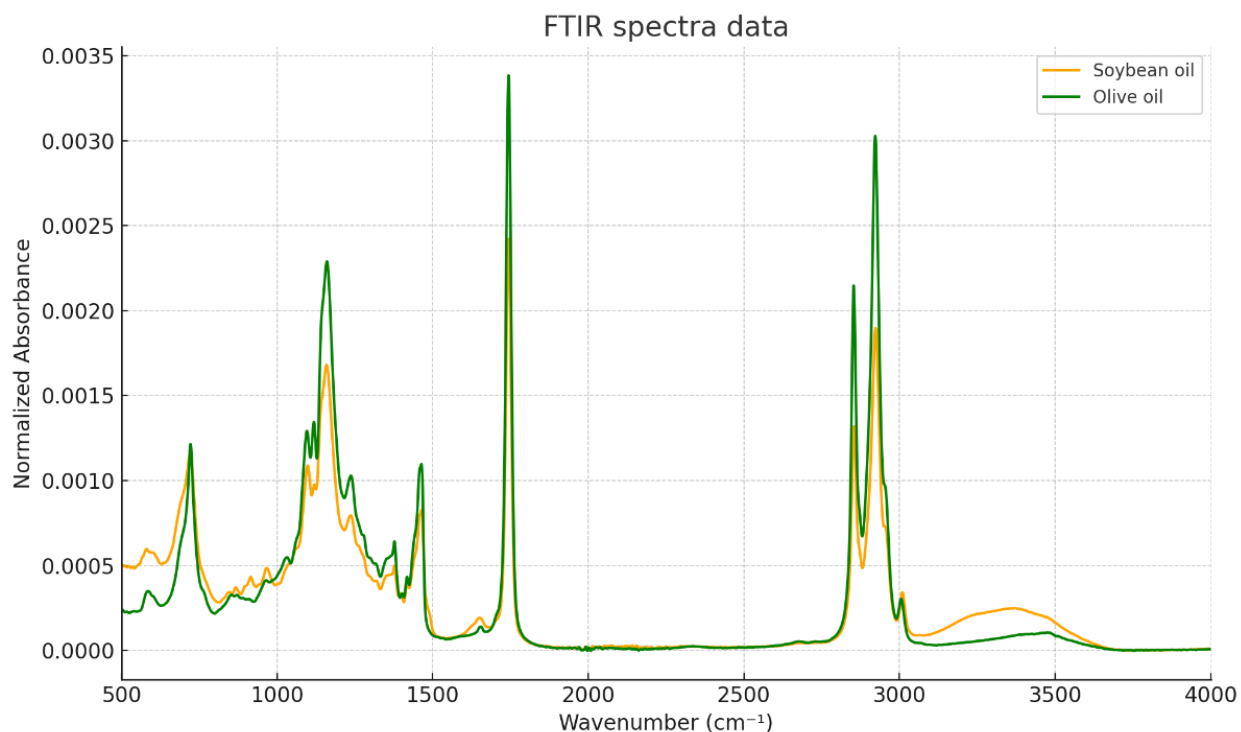


Figure 2.1: Spectra of different vegetable oils

Table 2.1 complements the information of the FTIR spectra and summarizes the main absorption bands and their relationship to thermal degradation processes in vegetable oils. The bands are identified by their respective wavenumbers (cm^{-1}) and include descriptions of the corresponding vibrations. Previous applications are also reported, demonstrating the usefulness of each band in differentiating fresh and degraded oils, as well as in the detection of compounds associated with oil degradation.

Table 2.1: Main absorption bands observed in FTIR applied to monitor thermal degradation in vegetable oils and its application in previous cases

Wavenumber	Description	Previous analysis application
722	CH ₂ rocking vibration.	Absorption decreases after heating process [6] and reveal significant differences between fresh, used and adulterated oils [15][16]

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Wavenumber	Description	Previous analysis application
965	Bending vibrations of C-H	Absorption increases after heating process [6]
987	Bending vibrations of conjugated C-H trans	Absorption increases after heating process [6].
1096	C-O stretching vibrations (ester groups)	Used to discriminate types of vegetable oils [17]
1119	C-O stretching vibrations (ester groups)	May be used to discriminate types of vegetable oils [17][6]
1160	C-O stretching vibrations (ester groups). Bending vibrations (CH ₂ groups). C-C stretch of carotenoids	Decrease in intensity because triglycerides break down into degradation products [18][17][8][19][20][4][21]. A slight shift in the band may occur due to structural changes.
1237	C-O stretching vibrations (ester groups). Bending vibrations (CH ₂ groups)	May indicate variation in fatty acid composition. Thermal or oxidative processes typically decrease the intensity of the band [6][22][23][24]
1377	Bending vibrations (CH ₂ groups)	Moderate decrease in intensity during thermal oxidation; associated with methylene deformation in aliphatic chains [22]
1400	Bending vibrations C-H (cis-olefinic groups)	Shifts to low wavenumbers but it becomes stronger as the content of polyunsaturated fatty acids (PUFA) increases [17]
1418	Rocking vibrations C-H	Decrease the intensity of the band. Methylene groups decrease due to the degradation of fatty acids and the formation of secondary oxidative compounds [6][22]
1464	Bending vibrations CH ₂ (aliphatic groups)	Decrease the intensity of the band particularly with the breakdown of aliphatic chains and the formation of smaller compounds [6][4][25][26]
1653	C=C stretching vibrations (cis olefins)	Decrease in intensity due to oxidation of unsaturated fatty acids [6][16][27]

Continued on next page...

Wavenumber	Description	Previous analysis application
1744	Stretching vibrations of triglycerides (Ester carbonyl C=O)	Distinctive carbonyl stretching mode of the ester linkages of triglycerides. Decrease in intensity due to the degradation of triglycerides and the formation of secondary compounds such as free fatty acids, aldehydes, and ketones [15][6][17][13][28][29]
2854	Symmetric stretching vibrations C-H Aliphatic CH ₂	Referred to methyl group -CH ₃ . Alterations in the chemical structure, such as the formation of oxygenated compounds, can modify the intensity or profile of the band [17][22][24][13][30]
2873	Symmetric stretching vibrations C-H Aliphatic CH ₃	Decrease in intensity [6][26]
2923	Antisymmetric stretching vibrations C-H Aliphatic CH ₂	Decrease in intensity due to the degradation of hydrocarbon chains, which reduces the number of intact methylene groups [17][16][8]
2956	Antisymmetric stretching vibrations C-H Aliphatic CH ₃	Decrease in intensity due to the degradation of triglycerides and the breakdown of hydrocarbon chains [6][4][23]
3007	Symmetric stretching vibrations C-H in double bonds C=C-H	Absorption decreases after heating process due to oxidation of unsaturated fatty acids. It is a direct marker of the level of unsaturation in oils [6][27]
3475	Stretching vibration of the ester group triglycerides – overtone	Presence of hydroxyl-containing compounds. Indicator of the formation of polar and oxidative compounds [6][31][32][33][34][35][36]

UV-Vis spectrophotometry complements FTIR by providing specific quantitative markers for conjugated oxidation products. The specific extinction coefficients K232 and K270 are among the most widely used UV-Vis parameters in oil quality assessment, reflecting the concentration of conjugated dienes and trienes formed during lipid oxidation, respectively [37]. K232 responds

primarily to primary oxidation products such as hydroperoxides, while K270 tracks secondary products including aldehydes and ketones that accumulate as degradation advances [15]. Together, these indices provide a staged picture of the oxidation process that complements the molecular vibration information captured by FTIR. For soybean oil, the UV-Vis range between 200 and 280 nm is particularly informative, as the progressive increase in both coefficients during thermal treatment directly reflects the consumption of polyunsaturated fatty acids and the accumulation of secondary oxidation compounds [37]. Zhang et al. [38] developed a quantitative method for monitoring acid value and heating-induced changes in edible oils by UV-Vis, confirming the sensitivity of these parameters. Jolayemi et al. [39] showed that UV-Vis spectra could also be applied to classify palm oils according to geographical origin, demonstrating the versatility of this technique beyond oxidative stability assessment.

The integration of FTIR and UV-Vis has been suggested as a robust approach to characterize oil degradation comprehensively. Meenu et al. [5] reviewed applications of vibrational spectroscopy in frying oils and emphasized that while FTIR captures molecular vibrations associated with oxidation, UV-Vis provides complementary information on conjugated systems and pigment degradation. Studies such as Jamwal et al. [3] further highlighted the relevance of combining both techniques with chemometric tools to extract maximum information from complex degradation pathways.

Used together, FTIR and UV-Vis spectroscopy provide sensitive markers for degradation, yet their full potential is realized when supported by data-driven analysis. The reliance on predefined bands or indices remains a limitation, underscoring the need for frameworks capable of systematically identifying the most responsive variables while maintaining chemical interpretability.

2.3 Chemometric Approaches in Spectroscopy

The complexity of FTIR and UV-Vis spectra, characterized by overlapping bands and high collinearity among variables, makes direct interpretation difficult. Chemometric methods address this challenge by transforming large datasets into a reduced number of informative components, allowing the extraction of meaningful patterns that reflect chemical changes in the samples [40]. These methods are essential for studying dynamic systems such as oil degradation, where subtle variations in absorbance can indicate molecular transformations that are not apparent in raw spectra [41].

Principal Component Analysis (PCA) is one of the most widely used unsupervised methods in spectroscopic data analysis. It reduces dimensionality by transforming correlated spectral variables into orthogonal components that capture the maximum variance of the dataset [8, 26, 42]. In oil analysis, PCA has been applied to identify clusters associated with degradation stages or adulteration levels, enabling visualization of temporal or compositional trends [21]. For example, Jamwal et al. [3] demonstrated the capacity of PCA to distinguish fresh and oxidized edible oils based on

FTIR spectral variance, while Valantina et al. [43] used PCA to evaluate thermal degradation of palm oil through viscosity and spectral data.

Supervised chemometric techniques such as Partial Least Squares Regression (PLSR) have proven effective for quantitative modeling. PLSR identifies latent variables that maximize covariance between spectral data and reference chemical measurements, such as peroxide value, anisidine value, or iodine value [8]. Several authors have reported strong correlations between FTIR absorbance patterns and oxidation indices using PLSR, achieving accurate prediction of chemical parameters without extensive sample preparation [44]. For instance, Han et al. [8] and Nascimento et al. [13] demonstrated the ability of PLSR models to quantify oxidation in vegetable oils using ATR-FTIR spectra.

Chemometric methods applied to spectroscopic data are well established in analytical chemistry literature. Wold et al. [45] introduced PCA and partial least squares as foundational tools for multivariate spectral analysis, and their application to food and lipid systems has since been extensively validated. Overall, chemometric analysis has transformed the interpretation of spectroscopic data by enabling objective quantification and classification. Yet, the effectiveness of these methods depends strongly on the selection of input variables. Most implementations still rely on empirically defined wavenumbers or spectral regions, which may obscure subtle but chemically relevant variations. Developing data-driven strategies for feature selection, such as slope-based frameworks, offering a pathway to increase reproducibility and interpretability in chemometric modeling of oil degradation.

2.4 Machine Learning for Spectroscopic Data

The integration of machine learning (ML) with spectroscopic analysis has emerged as a powerful approach to extract meaningful information from high-dimensional data. Unlike conventional chemometric methods that rely on linear relationships or low-dimensional projections, ML algorithms can model complex, non-linear interactions among spectral variables, improving classification and prediction accuracy [9]. These capabilities have positioned ML as a key analytical tool for quality assessment, authentication, and degradation monitoring in edible oils and other complex materials [11], [10].

In FTIR and UV-Vis spectroscopy, ML models can learn from large datasets to associate subtle spectral variations with chemical or physical properties. Supervised regression algorithms such as Partial Least Squares Regression (PLSR), Random Forest (RF), and Support Vector Regression (SVR) are frequently used for predicting quantitative indices such as acid value, peroxide value, and iodine value [27]. Random Forest algorithms have also shown promise due to their robustness to noise and ability to capture variable interactions without extensive preprocessing of datasets [27].

Unsupervised learning methods remain valuable for exploratory analysis and unsupervised classification. Their combination with ML pipelines allows visualization of degradation trajectories and automatic detection of sample groupings [46]. Such approaches have been successfully applied to edible oils to distinguish degradation stages, geographical origins, and degrees of adulteration. Siddiqui et al. [47] used Support Vector Machines to classify adulterated meat and oil mixtures, achieving high discrimination accuracy.

Deep learning models, such as Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN), have recently been explored for their capacity to learn hierarchical representations directly from raw spectra. Although these models provide excellent predictive performance, they often lack interpretability, which limits their direct application in analytical chemistry. Trevisan et al. [12] noted that maintaining chemical interpretability is essential for ensuring that ML outcomes remain scientifically meaningful and not merely statistical correlations.

Despite these advances, a persistent challenge lies in identifying which spectral variables carry the most relevant chemical information. Many ML models are applied to full spectra or empirically defined regions, which increases dimensionality and the risk of overfitting [10]. Data-driven feature selection strategies, such as the slope-based framework we developed, slope-based framework developed in this research, address this limitation by isolating wavenumbers that exhibit the highest temporal responsiveness during degradation. This approach not only improves model efficiency but also preserves the interpretability required for linking ML predictions to underlying chemical transformations [9].

Machine learning therefore complements and extends traditional chemometrics by providing flexible, scalable, and adaptive models for analyzing spectroscopic data. When combined with systematic feature selection, it enables the construction of predictive models that can generalize across different oils and experimental conditions, bridging the gap between spectroscopy and quantitative chemical analysis.

2.5 Gaps in Literature

Despite extensive research on oil degradation using spectroscopic and chemometric techniques, several methodological and interpretative limitations remain. Most existing studies depend on pre-defined wavenumber regions or empirical indices derived from conventional literature. Although such approaches have been effective for specific systems, they lack reproducibility when experimental conditions or oil compositions change. This dependence on expert selection introduces subjectivity and hinders the comparability of results across datasets and instruments.

Another critical gap concerns feature selection. In many studies, chemometric and machine learning models are applied directly to full spectra or broad regions without objective criteria for identifying

the most responsive variables. This increases dimensionality, introduces redundancy, and may dilute chemically meaningful information. Consequently, model interpretability and transferability are compromised, especially when extended to new samples or degradation environments.

While machine learning has significantly improved predictive performance, its adoption in oil degradation studies remains fragmented. Many models achieve high accuracy but fail to provide chemical insight into the specific vibrational or electronic transitions responsible for prediction. This disconnection between statistical performance and chemical understanding limits the practical adoption of ML-based frameworks in analytical chemistry.

There is therefore a need for a systematic, data-driven methodology capable of identifying dynamic and chemically relevant spectral regions without prior assumptions. Such an approach should combine interpretability with predictive accuracy, enabling both qualitative understanding of degradation mechanisms and quantitative prediction of chemical parameters. Our slope-based framework addresses this gap by quantifying spectral responsiveness through regression analysis and linking selected variables with machine learning models for predictive applications such as iodine value estimation.

Table 2.1 summarizes key representative studies on spectroscopic and chemometric analysis of oils, illustrating the progression of methodological developments and the specific gaps that motivate the present work. Table 2.1. Each study contributed to the advancement of spectroscopic analysis of oils, particularly using FTIR combined with chemometrics or machine learning. However, most relied on empirical or fixed-band selection strategies, lacked dynamic variable identification, or emphasized statistical performance without interpretability. Collectively, these limitations reveal a consistent need for a reproducible, data-driven framework capable of quantitatively identifying responsive spectral features and linking them to chemical behavior. This synthesis directly supports the motivation for developing the slope-based framework proposed in this thesis.

Information summarized shows that while chemometrics and machine learning have improved the statistical treatment of spectroscopic data, they still depend on expert judgment for variable selection. This reliance constrains the generalization of models and complicates the comparison between different datasets, instruments, or oil types. Beyond this, recent works have demonstrated that even advanced algorithms, such as support vector machines or neural networks, provide limited chemical insight when applied to unfiltered spectral data. These models may achieve accurate predictions but do not reveal which wavenumbers or vibrational modes drive the observed changes.

This opens a clear opportunity to establish a data-driven, reproducible framework for feature selection and model construction. A slope-based framework can address this gap by quantifying the rate of spectral change at each wavenumber and identifying the most responsive regions over

Table 2.2: Summary of representative studies on spectroscopic and chemometric analysis of oils and their relevance to the present work.

Reference	Main Contribution	Identified Limitation / Relevance to Present Work
Trevisan et al. (2012) [12]	Computational extraction of biological information from FTIR data using PCA and clustering.	Demonstrated the need for reproducible feature extraction; did not propose a quantitative variable selection method.
Jamwal et al. (2021) [3]	Detection of edible oil oxidation using FTIR and PCA.	Relied on fixed spectral regions; lacked generalization across different oils.
Valantina et al. (2022) [43]	Thermal degradation of palm oil evaluated by viscosity and FTIR analysis.	Focused on single-band monitoring; limited statistical treatment of spectra.
Cakmak-Arslan (2022) [6]	FTIR monitoring of hazelnut oil quality during thermal processing.	Used predefined bands; interpretability limited by empirical selection.
Han et al. (2020); Nascimento et al. (2021) [8]	Application of PLSR to correlate FTIR data with chemical indices.	Achieved accurate predictions but without systematic variable selection.
Poiana et al. (2015) [7]	LDA-based classification of olive oil adulteration using FTIR.	Effective classification; however, required manual identification of relevant regions.
Siddiqui et al. (2021) [47]	Multivariate SVM classification for detection of lard adulteration in meat using FTIR.	Applied advanced ML models; lack interpretability and dynamic feature extraction.
Deekshitha et al. (2024) [48]; Bai et al. (2022) [49]	Spectroscopic evaluation of oil treatments and sterol degradation.	Highlighted need for objective, generalizable methods like slope-based selection.

time. By integrating this method with machine learning models, it becomes possible to link spectral dynamics. Ultimately, we envision the slope-based framework evolving toward online monitoring systems, multimodal sensor fusion, and calibration transfer protocols that would make it directly deployable in industrial quality control environments with chemical parameters such as iodine value, providing both interpretability and predictive capability.

Our framework therefore aims to unify three methodological dimensions that have remained partially disconnected in previous studies: quantitative feature extraction from spectral time series, multivariate pattern recognition of degradation stages, and machine learning prediction of chemical indices. This integration defines the foundation of the present research and establishes the basis for the methodological design described in the following chapter.

Chapter 3

MATERIALS AND METHODS

3.1 Oil Samples and Thermal Degradation Protocols

We selected commercial soybean oil as the experimental material due to its high content of polyunsaturated fatty acids and its wide industrial and nutritional relevance. Its chemical composition, dominated by linoleic and linolenic acids, makes it particularly susceptible to oxidation and polymerization during heating, providing a suitable system for evaluating spectral indicators of degradation.

All experiments were conducted under controlled laboratory conditions to ensure reproducibility. Approximately three liters of commercial refined soybean oil were placed in electrical stainless-steel fryers and subjected to isothermal heating at 180 °C. The array is shown in Figure 3.1. This temperature was selected because it approximates the smoke point of soybean oil and represents a typical condition for domestic and industrial frying. The heating process was maintained for a total of sixty-four hours, allowing continuous degradation under stable conditions.

Mechanical agitation was applied throughout the experiment to maintain uniform temperature and prevent localized overheating. The temperature was monitored with a type-J thermocouple connected to a calibrated digital controller, ensuring stability within ± 1 °C. Oil samples of approximately 10 mL were collected at regular intervals every one hour, resulting in sixty-four samples. After collection, each sample was immediately transferred to amber glass vials with airtight caps to minimize exposure to light and oxygen. The samples were stored at room temperature in the dark until spectroscopic analysis.

This experimental configuration was designed to simulate realistic degradation conditions while maintaining analytical consistency. The controlled environment enabled the generation of a time-resolved dataset suitable for quantitative evaluation through FTIR spectroscopy, UV-Vis spectrophotometry, and subsequent chemometric and machine learning analysis.

3.2 Reference Chemical Assays

The chemical characterization of the thermally degraded soybean oil included the determination of its iodine value (IV). These parameters were selected because they are widely accepted indicators of oxidation and unsaturation, providing a direct link between spectroscopic observations and chemical transformations.



Figure 3.1: Array in the electrical fryer with thermocouples.

3.2.1 Iodine Value

The iodine value (IV) is a fundamental parameter for evaluating the degree of unsaturation in vegetable oils and their oxidative stability. It reflects the number of double bonds available for halogen addition and is expressed as grams of iodine absorbed per one hundred grams of oil. Oils with higher iodine values contain greater proportions of unsaturated fatty acids and are consequently more prone to oxidation, while those with lower values exhibit higher thermal and oxidative stability [50]. For soybean oil, typical IV values range between 120 and 140 g I₂ / 100 g, depending on its degree of refinement and fatty acid composition [51]. This parameter serves as a key indicator of degradation during prolonged thermal exposure, complementing spectroscopic measurements that monitor molecular transformations in the oil matrix [52], [53]. These methods rely on the Wijs titration principle, in which iodine monochloride reacts with the carbon–carbon double bonds of unsaturated fatty acids, and the unreacted reagent is subsequently back-titrated with standardized sodium thiosulfate. The reaction stoichiometry allows direct quantification of total unsaturation. Despite its robustness, the Wijs method is known to be time-consuming and requires careful control of reagent standardization and light exposure to ensure reproducibility [54].

We chose the iodine value as the principal chemical index to quantify the loss of unsaturation and to validate the spectroscopic and machine-learning analyses. The determination was carried out using the spectrophotometric method proposed by Soares and Rocha, which represents a rapid and environmentally friendly alternative to the conventional Wijs titration. This procedure, shown in Figure 3.2, is based on the discoloration of an aqueous triiodide solution following halogenation of unsaturated compounds in the oil. The remaining fraction of triiodide in the aqueous phase is quantified by UV–Vis spectrophotometry at 450 nm, and the iodine value is calculated from the inverse proportionality between absorbance and the degree of unsaturation.

The quantitative relationship is given by Equation 3.1, defined by Soares and Rocha.

$$IV = a - bA_{450} \quad (3.1)$$

where IV is the iodine value (g I₂ per 100 g of oil), A_{450} is the absorbance of the aqueous phase at 450 nm, while a and b are calibration coefficients determined experimentally using reference samples of known iodine value. The inverse proportionality between A_{450} and IV arises because the remaining triiodide concentration decreases as the iodine consumed in the halogenation reaction increases.

This method consumes only milliliter-scale volumes of reagents and generates minimal chemical waste while maintaining analytical equivalence to official reference procedures [55].

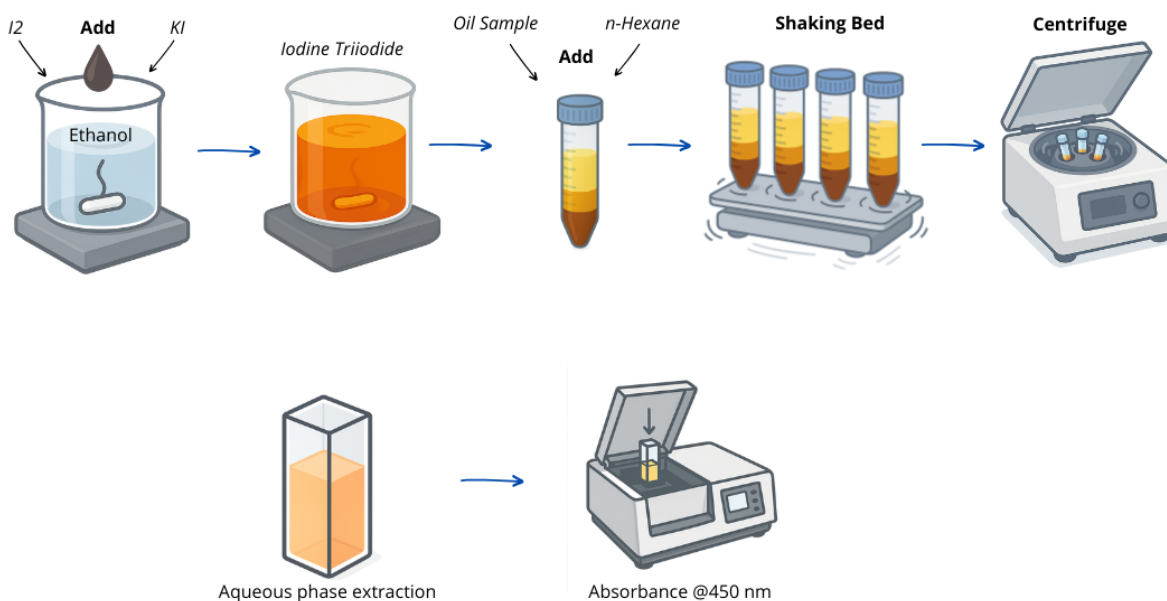


Figure 3.2: Spectroscopy method for IV determination

The selection of this method was motivated by its demonstrated accuracy, reproducibility, and operational simplicity. Soares and Rocha [55] verified that the spectrophotometric results were statistically equivalent to those obtained using the American Oil Chemists' Society (AOCS) Cd 1c-85 and DIN EN 14111:2003 reference methods at the 95 % confidence level, while requiring less than one-tenth of the reagent volume. Its analytical performance was later confirmed by independent studies [54, 56, 57], which successfully applied spectroscopic and chromatographic adaptations of the same principle to a variety of vegetable oils and biodiesel matrices [54, 58]. These developments collectively reinforce the reliability of the Soares method as a validated and sustainable procedure for routine iodine value determination.

3.2.2 UV–Vis Absorbance Indices (K232 and K270)

Ultraviolet–visible (UV–Vis) spectrophotometry was used to quantify the formation of conjugated oxidation products in thermally degraded soybean oil. The specific extinction coefficients K232 and K270 are well-established analytical parameters that reflect the concentration of conjugated dienes and trienes, respectively. These compounds arise from the rearrangement of double bonds during the oxidation of polyunsaturated fatty acids and serve as sensitive indicators of both primary and secondary oxidation processes [15].

The procedure followed the IUPAC 2.505 (1992) and AOCS Ch 5-91 official methods, with adaptations from De Leonardis et al. (2016) and Tan et al. (2022) for modern spectrophotometric acquisition [37], [15]. Oil samples were diluted to 1 % (w/v) in spectroscopic-grade isooctane and analyzed using quartz cuvettes of 1-cm optical path length. Absorbance spectra were recorded

between 200 and 400 nm, and the specific extinction coefficients were calculated according to Equation 3.2.

$$K_{\lambda} = \frac{A_{\lambda}}{c \times L} \quad (3.2)$$

where K_{λ} is the specific extinction coefficient at wavelength λ (232 or 270 nm), A_{λ} is the measured absorbance, c is the concentration of the oil in g / 100 mL of solvent, and L is the optical path length in cm. The values were expressed as K232 and K270, corresponding to absorbance at 232 nm and 270 nm, respectively.

The K232 index quantifies primary oxidation products such as hydroperoxides and conjugated dienes generated during the initial stages of lipid oxidation. In contrast, K270 reflects secondary oxidation products, including aldehydes and ketones, which accumulate after decomposition of hydroperoxides [37]. The increase of both indices with heating time or oxidation severity is indicative of progressive degradation and unsaturation loss. De Leonardis et al. (2016) demonstrated that refining and de-acidification significantly decrease both K232 and K270 values, confirming their sensitivity to oxidative history. Similarly, Tan et al. (2022) reported a direct, linear increase in K232 and K270 with higher adulteration or reuse levels of palm olein, validating their diagnostic value as universal oxidation markers. Their findings also supported that K270 tends to exhibit a larger magnitude than K232 because secondary oxidation products are more stable than hydroperoxides.

Here, the determination of K232 and K270 served a dual purpose: first, to complement the iodine value as an independent chemical validation of unsaturation loss, and second, to provide spectral indicators for cross-correlation with the FTIR-derived wavenumber dynamics.

The interpretation of K232 and K270 values provides insights into the oxidative stage of oil samples. Low values of K232 (< 2.0) and K270 (< 0.3) denote fresh oil with negligible oxidation, whereas increasing coefficients indicate the accumulation of conjugated systems and secondary degradation products. For heavily oxidized oils, K232 may rise above 4 and K270 above 1.0, reflecting advanced deterioration [37]. These parameters therefore act as complementary chemical references, confirming the molecular transformations detected by FTIR analysis and reinforcing the spectroscopic-chemometric integration central to this study.

3.3 Optical Spectroscopic Data Acquisition

3.3.1 ATR-FTIR

All spectra were recorded using a PerkinElmer Spectrum Two spectrometer equipped with an attenuated total reflectance (ATR) accessory featuring a diamond crystal and a ZnSe optical interface. The use of ATR minimized sample preparation and ensured reproducibility between measurements

by maintaining a constant optical path length and effective penetration depth within the mid-infrared region.

Each oil sample was directly placed on the ATR crystal using a pipette, covering the entire contact area to guarantee optimal coupling and eliminate air gaps. Before each measurement, the crystal was cleaned to prevent cross-contamination between samples. Background spectra were collected prior to every measurement session under the same instrumental conditions to correct for ambient atmospheric interference, primarily due to water vapor and CO₂.

Spectral acquisition was performed in the range of 4000–400 cm⁻¹, with a resolution of 4 cm⁻¹ and sixteen co-added scans per spectrum to improve the signal-to-noise ratio without compromising acquisition speed. Each sample spectrum was recorded at room temperature immediately after collecting from the thermal degradation system to minimize secondary oxidation.

The raw spectra were exported in absorbance format and subsequently processed following a standardized pipeline.

3.3.2 UV-Vis

All measurements were performed using a PerkinElmer Lambda 365 dual-beam spectrophotometer equipped with quartz cuvettes of 1 cm optical path length. The instrument was calibrated prior to each session using a reference blank of spectroscopic-grade isooctane, which also served as the solvent for sample preparation.

Each oil sample was diluted to 1 % (w/v) in isooctane to ensure linearity between absorbance and concentration according to the Beer–Lambert law. Absorbance spectra were acquired over the range 200–700 nm with a data interval of 1 nm, scan speed of 60 nm·min⁻¹, and a spectral bandwidth of 1.0 nm. The baseline was automatically corrected by subtracting the reference spectrum of pure isooctane from each measurement.

After acquisition, absorbance data were converted into numerical matrices for chemometric and slope-based processing. The reproducibility of measurements was verified by performing duplicate scans for each sample; the relative deviation between replicates did not exceed 2 %, confirming instrumental stability.

3.4 Data Preprocessing Pipeline

Baseline correction is one of the most applied preprocessing techniques in FTIR studies, addressing variations in spectral intensity caused by instrumental fluctuations or sample inconsistencies. While baseline correction improves consistency across spectra, it requires careful parameter selection to avoid over-correcting spectra and distorting peak intensities, which can affect multivariate analysis outcomes.

Smoothing and noise reduction techniques, such as Savitzky-Golay filtering and wavelet transformation, enhance spectral clarity by reducing random fluctuations. These methods are particularly useful in detecting subtle changes in oxidation markers, such as the formation of carbonyl groups (1744 cm^{-1}) and trans-isomers (966 cm^{-1}) in thermally degraded oils [59], [60]. However, excessive smoothing can remove important spectral details or alter peak shapes, leading to the loss of valuable chemical information.

Normalization and standardization are essential for correcting differences in sample concentration, path length, and light scattering [61], [62], [63]. Techniques such as vector normalization and min-max scaling ensure that spectral intensities remain comparable across multiple samples. Despite its advantages, normalization may mask actual intensity differences related to oxidation progression, especially in studies where absolute spectral intensity is an important marker of chemical degradation.

Derivative transformations (first and second order) are used to enhance peak resolution and minimize baseline effects, making overlapping spectral features more distinguishable. First-order derivatives highlight peak shifts [64], [32], while second-order derivatives emphasize subtle changes in spectral curvature, but requires careful optimization, as higher-order derivatives can amplify noise and introduce artificial spectral features, which may mislead multivariate analyses [59, 65, 66].

Wavenumber selection (feature reduction) is crucial for focusing on chemically relevant spectral regions while removing redundant or non-informative data [66, 67]. Most studies prioritize regions associated with oxidation markers (3006 cm^{-1} , 1744 cm^{-1} , 966 cm^{-1}) while excluding regions dominated by background noise or non-essential molecular vibrations. It is important to choose marker bands to avoid losing critical chemical information in post treatments like clustering and classification models [68, 69].

Preprocessing techniques significantly impact the quality of FTIR-based oil degradation analysis by enhancing spectral clarity, reducing variability, and improving model robustness. However, each method has trade-offs: overcorrection can distort true spectral patterns, excessive smoothing can remove valuable information, and improper normalization can obscure real differences between samples. The choice of preprocessing strategy must be carefully tailored to the research objective, ensuring that it enhances rather than biases spectral analysis.

Table 3.1 summarizes the key pre-treatment methods commonly applied in FTIR analysis, detailing their specific use cases and methodologies. It highlights how baseline correction, smoothing, normalization, derivative transformations, and wavenumber selection contribute to refining spectral quality.

The implementation of pre-processing techniques in FTIR spectroscopy is a critical step in ensuring

Table 3.1: Summary of key pre-treatment methods commonly applied in FTIR analysis

Pre-treatment	Application Cases	
Baseline Correction	Exhibit baseline drifts due to instrumental effects or sample inconsistencies. Methods like rubber band correction and multiplicative scatter correction (MSC) are used to adjust.	Herculano et al. (2021) [19]
Smoothing (Noise Reduction)	Enhance signal clarity without altering the main spectral features, especially in low-intensity regions. Used techniques include Savitzky-Golay filtering or wavelet transformation.	Valantina et al. (2022) [43]
Normalization and Standardization	Variability in sample concentration and path length that distort spectral intensities. Min-max normalization or vector normalization are applied.	Nascimento et al. (2021) [13]
Derivatives (First and Second Order)	Applied in overlapping cases in peak resolution. Make small spectral differences detectable.	Poiana et al. (2020) [7]
Wavenumber Selection	Optimizes computational efficiency by focusing on relevant spectral features. Reducing data dimensionality	Cakmak-Arslan et al. (2022) [6]

reliable spectral interpretation. These pre-treatment strategies not only improve the signal-to-noise ratio but also refine spectral patterns, making it easier to extract meaningful insights from FTIR data. Once the spectral data has been properly processed, advanced FTIR analysis techniques can be applied to extract quantitative and qualitative information about molecular transformations during oil degradation.

3.5 Slope-Based Variable Selection Framework

We developed the slope-based framework to identify the most dynamic and chemically informative wavenumbers within time-resolved FTIR datasets. Our method quantifies spectral reactivity through the rate of absorbance change over time, providing a data-driven alternative to empirical band selection. All calculations were performed in Python (NumPy, Pandas, and SciPy), and the full implementation is described below.

3.5.1 Linear Regression of Absorbance vs. Time

For each wavenumber in the preprocessed spectral matrix $X_{(n \times m)}$, a simple linear regression model was fitted between absorbance (A) and time (t). The slope (b_i) of the regression line represents the rate of change in absorbance intensity, reflecting the magnitude of molecular transformation at that wavenumber and is represented by Equation 3.3. The absolute value of the slope ($|b_i|$) was used as a quantitative descriptor of dynamic behavior.

$$b_i = \frac{\sum_{t=1}^n (t - \bar{t})(A_{i,t} - \bar{A}_i)}{\sum_{t=1}^n (t - \bar{t})^2} \quad (3.3)$$

where $A_{i,t}$ is the absorbance at wavenumber i and time t , $\bar{A}_i$ is the mean absorbance for wavenumber i , and $\bar{t}$ is the mean acquisition time. The resulting vector of slopes represents the global dynamic behavior of the spectrum across the entire degradation period.

This linear relationship assumes that the absorbance changes approximately monotonically with degradation time, which is valid for most chemical processes under controlled heating conditions. The slope thus serves as an intrinsic kinetic descriptor derived directly from spectral evolution.

3.5.2 Segment-Based Selection (100 cm^{-1} bins)

To ensure uniform spectral coverage and avoid redundancy between adjacent variables, the entire FTIR range (4000–400 cm^{-1}) was divided into 36 segments of 100 cm^{-1} each. Within each segment, the wavenumber showing the maximum absolute slope ($|b_i|_{\max}$) was selected as the representative variable for that region. This segmentation balances spatial resolution and interpretability, producing a compact set of candidate features distributed evenly across the spectrum.

This procedure enhances reproducibility because each segment yields one independent variable selected by quantitative behavior rather than visual inspection or literature-defined bands. The approach is fully data-driven and consistent across datasets, allowing cross-system comparisons.

3.5.3 Extraction of Top Dynamic Wavenumbers

Once representative variables were identified from each 100 cm^{-1} segment, their slopes were ranked in descending order. The five wavenumbers with the highest absolute slopes were defined as the most dynamic wavenumbers. These wavenumbers correspond to the regions exhibiting the strongest absorbance variation during thermal degradation and are potentially the most chemically responsive indicators of molecular change.

The five most dynamic wavenumbers identified were 1731, 988, 709, 699, and 1112 cm^{-1} , and were later used for kinetic interpretation and visualization of degradation trajectories. The band at 1731 cm^{-1} corresponds to C=O stretching of ester carbonyl groups in triglycerides and oxidation-

derived carbonyl compounds; its systematic change during heating reflects both the breakdown of triglyceride esters and the accumulation of aldehydic and ketonic secondary products [3, 6]. The band at 988 cm^{-1} is associated with C-H bending vibrations of conjugated trans double bonds formed during oxidation-induced geometric isomerization of fatty acids, making it a sensitive marker of unsaturation rearrangement [6]. The bands at 709 and 699 cm^{-1} correspond to CH_2 rocking vibrations of long aliphatic hydrocarbon chains; their combined response reflects structural changes in the fatty acid backbone associated with chain degradation and the formation of shorter-chain products [16]. Finally, 1112 cm^{-1} falls within the C-O stretching region of ester linkages in triglycerides, sensitive to hydrolytic cleavage and the formation of free fatty acids during prolonged heating [13]. The consistent identification of these five wavenumbers across replicate degradation sequences confirms that their dynamic behavior reflects genuine chemical transformation rather than instrumental noise. These features form the foundation of the chemometric, and regression analyses described in the subsequent sections.

To provide a structured overview of the analytical procedure, the complete FTIR data processing workflow is summarized in Figure 3.3. The schematic highlights each stage in the pipeline, including spectral acquisition, preprocessing, slope-based feature extraction, and multivariate analysis. This modular and reproducible workflow enables interpretable, high-resolution monitoring of degradation and can be adapted to other time-resolved spectroscopic applications.

3.6 Machine Learning and Multivariate Analysis

We applied machine learning (ML) and multivariate statistical methods to evaluate the representativeness and predictive capacity of the slope-based spectral variables. Our analysis workflow was implemented entirely in Python 3.11 using the scikit-learn library (v 1.3.2) together with NumPy, Pandas, and Matplotlib. All routines were executed in a controlled environment to ensure reproducibility and transparency of data processing.

The analytical strategy followed a hierarchical structure. First, we applied unsupervised techniques to explore intrinsic trends and assess the temporal coherence of the degradation process. Then, we developed supervised regression models to establish quantitative relationships between spectral descriptors and the iodine value. This combined approach allowed the same dataset to be interpreted both qualitatively, in terms of molecular evolution, and quantitatively, in terms of predictive power.

The prediction task maps a compact set of FTIR descriptors to iodine value measured by the spectrophotometric procedure described in Section 3.2. The dataset contained one hundred and eight paired observations assembled from two sequences of fifty-four samples each. Data was partitioned into sixty-four rows for training, twenty-two rows for validation, and twenty-two rows for testing. The same preprocessing and evaluation pipeline was applied to every model to ensure

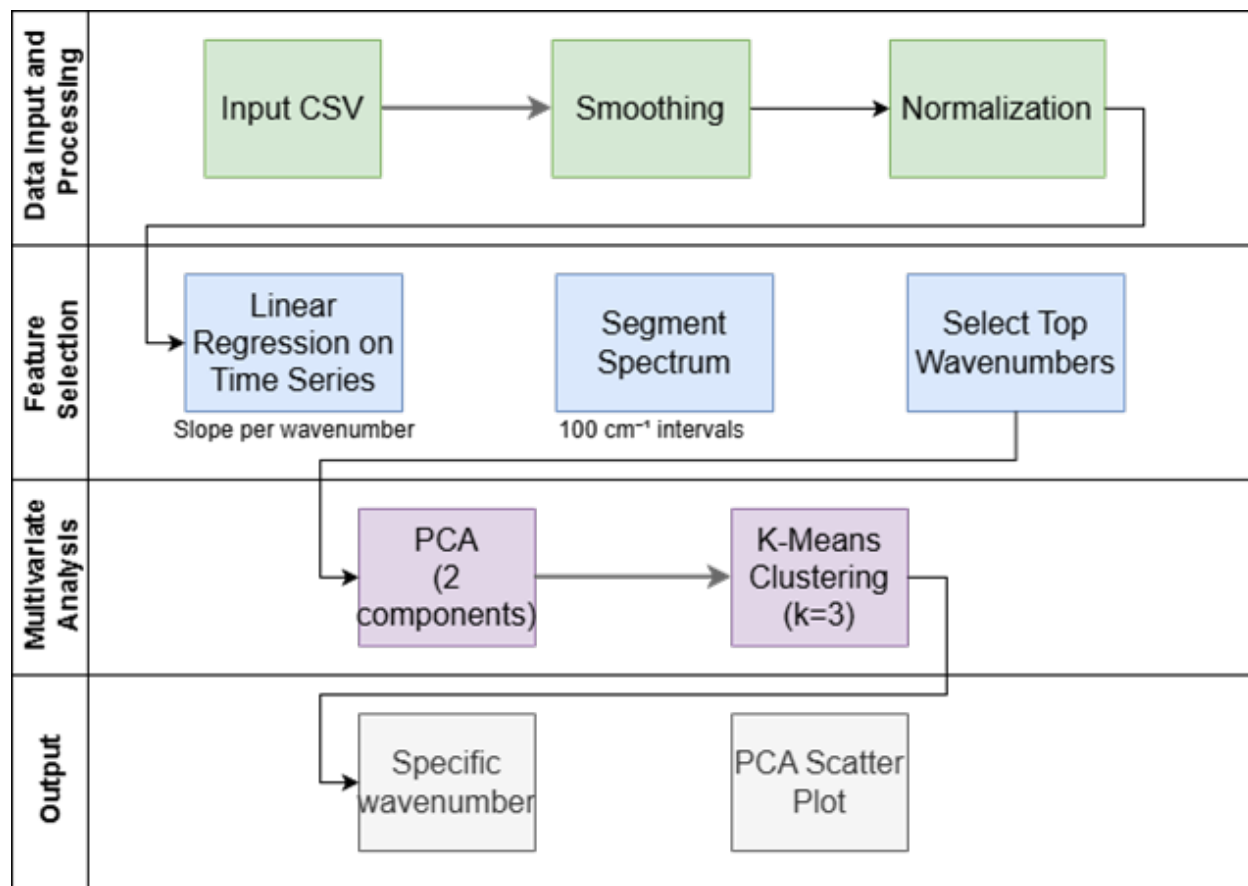


Figure 3.3: Data analysis workflow for FTIR spectroscopy.

fair comparison and full reproducibility. Predictors were centered and scaled within each training fold only, and the learned transformation was applied to validation and test data. All random seeds, software versions, and grid specifications were recorded.

Feature construction followed the slope-based framework. For each time point, the normalized absorbance at the selected dynamic wavenumbers served as predictors. The response variable was the iodine value measured on the same sample. No imputation was required because the pairs were completed after integrity checks. Collinearity diagnostics were computed to characterize the descriptor set, and residual analyses were used later to assess calibration.

Model selection covered linear baselines and tree ensembles to balance transparency and flexibility. Hyperparameters were tuned with grid search inside a nested cross-validation (CV) scheme that preserved the training and validation split. Final numbers reported in Section 5.4 correspond to the held-out test set that was not used during tuning.

3.6.1 Principal Component Analysis (PCA)

Principal Component Analysis was used as the initial exploratory tool to visualize the global variance structure of the slope-selected variables and to verify that they captured the main chemical dynamics of thermal degradation. The preprocessed spectral matrix $X_{(n \times m)}$ was mean-centered, and PCA was applied through eigenvalue decomposition of the covariance matrix expressed in Equation 3.4:

$$X = TP^T + E \quad (3.4)$$

where T is the scores matrix representing the projection of the samples in the new orthogonal space, P is the loadings matrix defining the contribution of each variable to the components, and E is the residual matrix.

3.6.2 K-means Clustering

K-means clustering was employed to classify samples according to spectral similarity and to identify distinct degradation stages. The number of clusters k was set to three based on two convergent criteria: the elbow method applied to the within-cluster inertia curve, which showed a marked reduction in improvement beyond $k=3$, and silhouette analysis, which yielded the highest mean coefficient at $k=3$ across repeated subsampling runs. This choice is also consistent with the expected chemical progression of thermal degradation, which literature characterizes as having three broad stages: an induction period with moderate oxidation, an active degradation phase with rapid unsaturation loss, and an advanced stage dominated by secondary oxidation products [1, 5]. The algorithm partitions the dataset into k clusters by minimizing the within-cluster sum of squared distances using Equation 3.5:

$$J = \sum_{j=1}^k \sum_{i \in C_j} \|x_i - \mu_j\|^2 \quad (3.5)$$

where x_i denotes a sample vector, μ_j is the centroid of cluster j , and C_j is the set of samples belonging to that cluster. Cluster quality was evaluated using the silhouette coefficient, which quantifies both cohesion and separation of clusters, and by the inertia metric reflecting total within-cluster variance.

3.6.3 Linear and Ridge Regression

Multiple linear regression provided a transparent baseline to quantify direct linear relations between each spectral marker and iodine value. Estimation used ordinary least squares with an intercept after standardization. Variance inflation factors and condition indices were examined to detect

collinearity among predictors. Ridge regression was included as a stabilized variant when mild dependence appeared among variables. The regularization parameter was chosen by inner cross validation. These baselines establish a lower bound of accuracy and yield coefficients with clear chemical interpretation.

3.6.4 Random Forest and Extra Trees

Random Forest and Extremely Randomized Trees were trained to model nonlinear relations and interactions among spectral markers without feature engineering. The number of trees, maximum depth, minimum samples per split, and minimum samples per leaf were tuned by grid search. Out of bag estimates were retained as an internal generalization check. Variable importance was computed using impurity-based indices and verified with permutation importance to ensure stability of the rankings. Partial dependence curves were inspected to visualize the functional effect of influential wavenumbers while marginalizing the rest.

3.6.5 Gradient Boosting

Gradient Boosting Regression was used to build additive ensembles of shallow trees optimized stage by stage. Learning rate, number of estimators, maximum depth, and subsampling ratio were tuned within the inner cross validation loop. Early stopping on the validation split prevented overfitting when convergence occurred before the maximum number of stages. Residual plots were reviewed to check calibration at low and high iodine values.

3.6.6 XGBoost

Extreme Gradient Boosting was evaluated as a high-capacity learner with regularization and shrinkage. The search space included number of estimators, learning rate, maximum depth, minimum child weight, gamma for split regularization, and subsampling of rows and features. Calibration was examined with parity plots on the held-out test set and uncertainty was summarized by the dispersion of cross validated errors.

This modeling strategy provides a coherent spectrum from fully linear to flexible ensemble learners while keeping a single, reproducible pipeline. Section 5.4 presents parity plots, cross-validated metrics, and comparative results for all models trained under this protocol. Performance metrics and validation criteria for all models described above are detailed in Section 3.7.

3.7 Validation and Metrics

Validation was conducted to ensure the reliability and reproducibility of both the unsupervised and supervised analyses derived from the slope-based framework. Validation strategies were designed to quantify the consistency of clustering, the robustness of dimensionality reduction,

and the predictive accuracy of regression models. All performance metrics were computed using Python's scikit-learn library, ensuring standardized statistical definitions and comparability across experiments.

3.7.1 Cross-Validation Strategy

For all regression models, validation followed a k-fold cross-validation protocol with $k = 10$, in accordance with standard practices in chemometrics and machine learning. The dataset was partitioned into ten equal subsets; nine were used for model training and one for validation in each iteration. The process was repeated until each subset had served once as the validation fold.

This strategy allowed robust estimation of model generalization by averaging performance metrics across folds, mitigating bias introduced by random partitioning. To prevent overfitting, hyperparameters were optimized exclusively within the training folds through nested cross-validation, while the outer loop was reserved for unbiased model evaluation.

For unsupervised methods such as PCA and K-means clustering, validation was performed through resampling stability tests, in which random subsets of the dataset (80 % of samples) were repeatedly analyzed to assess the consistency of component loadings, cluster assignments, and silhouette scores. This ensured that the extracted structure was intrinsic to the data and not an artifact of sampling noise.

3.7.2 Explained Variance and Dimensionality Assessment

The performance of PCA was evaluated by calculating the explained variance ratio (EVR) for each principal component, defined by Equation 3.6:

$$EVR_j = \frac{\lambda_j}{\sum_{i=1}^p \lambda_i} \quad (3.6)$$

where λ_j represents the eigenvalue of component j , and p is the total number of retained components. The cumulative explained variance provided a quantitative measure of how effectively the slope-based variables captured the chemical variability of the dataset.

3.7.3 Clustering Metrics

The performance of K-means clustering was evaluated using two complementary measures: silhouette coefficient and inertia.

The silhouette coefficient (s) was computed in Equation 3.7:

$$s = \frac{b - a}{\max(a, b)} \quad (3.7)$$

where a is the mean intra-cluster distance of a sample and b is the mean distance to the nearest cluster. Silhouette values range from -1 to 1 , with higher scores indicating better-defined clusters.

The inertia metric, defined as the within-cluster sum of squared distances, was used to measure cluster compactness. Lower inertia values indicated tighter grouping, which, in combination with high silhouette scores, validated the structural stability of the clustering.

3.7.4 Regression Model Performance Metrics

Model accuracy and generalization were quantified using the coefficient of determination, the root mean square error (RMSE) and the mean absolute error (MAE). These metrics were computed on the validation folds during tuning and on the held-out test set for final reporting. The coefficient of determination measures the proportion of variance in the reference iodine values explained by the model and is defined by Equation 3.8.

$$R^2 \quad (3.8)$$

where y_i are measured iodine values, $\bar{y}_i$ are predictions, and $\bar{y}$ is the sample mean. Values closer to one indicate stronger explanatory power.

The root mean square error summarizes the typical magnitude of prediction errors in the original units of iodine value and is defined by Equation 3.9

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (3.9)$$

Lower values indicate better accuracy, and the metric is sensitive to large deviations.

The mean absolute error provides a robust measure of average error magnitude and is defined by Equation 3.10:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \bar{y}_i| \quad (3.10)$$

MAE is less influenced by outliers than RMSE and complements it in practice.

To estimate generalization reliably, models were tuned with nested cross validation. Inner folds were used for hyperparameter search, while outer folds provided unbiased validation scores. The $2R^2$ in Section 5.4 reports the mean and the standard deviation of R^2 across cross validation folds as CV R^2 mean and CV R^2 standard deviation. A higher CV R^2 mean indicates stronger expected

performance on unseen data and a smaller CV R^2 standard deviation indicates stability of that performance across resamples.

Train, validation, and test partitions were kept disjoint to prevent information leakage. Training metrics reflect fit to the learning data. Validation metrics guided model selection during development. Test metrics quantify final performance on data not used at any stage of tuning. Parity plots and residual analyses complemented these metrics by inspecting calibration against the identity line, error symmetry, and variance homogeneity. Where relevant, observed errors were interpreted relative to the analytical uncertainty of the iodine value procedure described in Section 3.2, which provides a practical lower bound for RMSE and MAE.

Chapter 4

METHODOLOGICAL DEVELOPMENT AND SPECTROSCOPIC ANALYSIS OF VEGETABLE OIL DEGRADATION

4.1 Identification of Dynamic FTIR Wavenumbers

We used the thermally degraded soybean oil dataset as the primary benchmark for validating our slope-based feature extraction methodology.

To evaluate the performance of our method, we applied two comparison strategies to the soybean oil dataset: (i) a set of user-defined wavenumbers frequently reported in the literature for monitoring oil degradation (e.g., 722 [6], [15, 16], 987 [6], 1160 [70], and 3007 cm^{-1} [6], and (ii) the integration of absorbance in predefined spectral regions (area under the curve, AUC) *frequently associated with oxidation and compositional change* [71–73], in this case we use 3100–2900, 1800–1700, 1500–1400, 1200–1100 cm^{-1} .

The absolute value of the slope was used as a quantitative approximation for spectral responsiveness, to identify the rate of change associated with chemical transformations during thermal degradation. This approach enables objective identification of the most dynamic and chemically informative regions in the spectrum (Figure 4.1), without relying on prior assumptions or empirical selection.

Our framework identified the most dynamic wavenumbers, and these are the following dynamic wavenumbers: 1731, 988, 709, 699, and 1112 cm^{-1} . The behavior of selected wavenumbers is illustrated in Figure 4.2, as those exhibiting the highest temporal variation during the degradation process. These wavenumbers correspond to well-identified molecular vibrations associated with oxidation and structural changes in fatty acids and triglycerides, including carbonyl stretching, C–H deformation, and C–O stretching. Linear regression of normalized absorbance over time confirmed a consistent and monotonic trend in each of the selected wavenumbers, supporting their suitability as degradation indicators.

4.2 Comparison of Variable Selection Strategies

We evaluated the slope-based framework against two conventional approaches commonly used in spectroscopic data interpretation: the literature-defined band selection method and the integrated area-under-the-curve (AUC) strategy. This comparison aimed to determine whether the slope-based framework improved reproducibility, interpretability, and sensitivity in detecting spectral changes associated with oil degradation.

The literature-defined band approach relies on selecting determined spectral regions previously

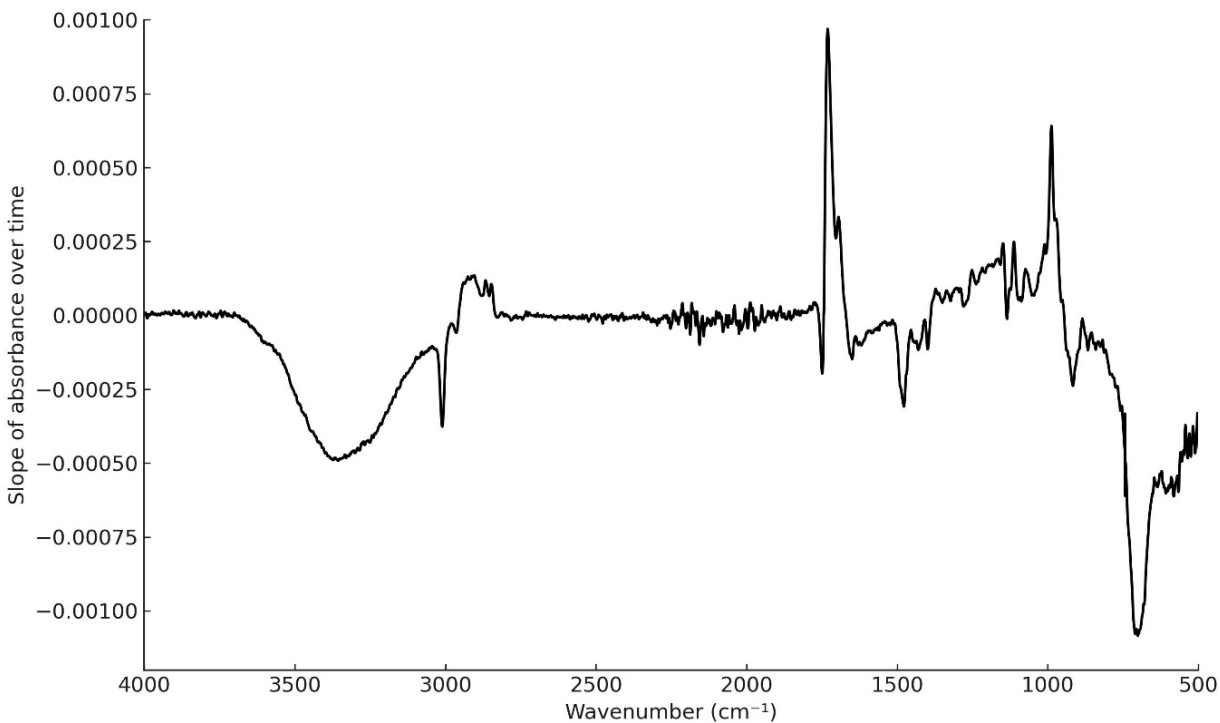


Figure 4.1: Slope per wavenumber in time. Linear regression was applied to obtain the behaviour of the FTIR data.

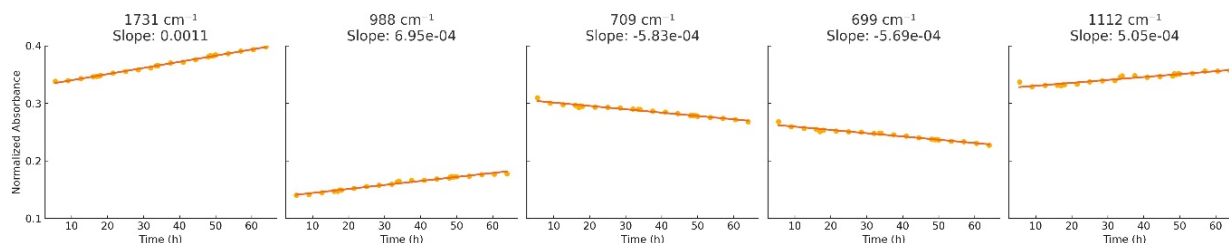


Figure 4.2: Linear regression of normalized absorbance at selected dynamic wavenumbers.

reported as oxidation markers in vegetable oils. These typically include the carbonyl band near 1744 cm^{-1} , the unsaturation band around 3007 cm^{-1} , and the aliphatic CH stretching region between 2924 and 2854 cm^{-1} . Although these bands have a well-established chemical significance, their selection is inherently empirical and may vary depending on instrument configuration, oil composition, or experimental conditions. When this approach was applied to the present dataset, the observed absorbance variations correctly reflected oxidation trends but showed limited sensitivity to subtle molecular transformations occurring outside these narrow regions.

The second approach, based on AUC integration, considers the cumulative absorbance within broader intervals of the infrared spectrum. We integrated representative ranges of $1800\text{--}1600\text{ cm}^{-1}$, $1300\text{--}1000\text{ cm}^{-1}$, and $800\text{--}650\text{ cm}^{-1}$ following previously published methodologies. This

method effectively smoothed local spectral noise and captured general changes in functional-group absorbance. However, it also averaged out localized variations and reduced the precision of wavenumber-level analysis. While AUC integration provided a global indicator of oxidation, it lacked the granularity necessary to identify specific vibrational modes with high temporal responsiveness.

In contrast, the slope-based framework offers a quantitative and objective alternative by computing the rate of change in absorbance at each wavenumber independently. The resulting slope distribution contains both direction and magnitude of change, providing a direct measure of spectral reactivity. When compared with literature-defined and AUC-based variables, our slope-derived features showed stronger alignment with the temporal evolution of the iodine value, as evidenced by the superior PCA trajectory ordering and clustering coherence described in Sections 4.3 and 4.4. This result also illustrates how our approach better captures the dynamic chemical information coefficients with reference chemical indices such as the iodine value and UV–Vis absorbance parameters K232 (specific extinction at 232 nm) and K270 (specific extinction at 270 nm). This demonstrated that the slope-based selection not only identified chemically relevant regions but also preserved the dynamic information lost in averaged or empirically chosen bands.

Another important distinction among the three methods lies in their reproducibility and transferability. Literature-based band selection depends heavily on prior assumptions and is sensitive to small instrumental shifts, whereas AUC integration requires manual definition of interval boundaries that may not generalize across datasets. The slope-based framework, by contrast, is fully data-driven and independent of prior spectral knowledge. Its reliance on numerical regression rather than fixed wavelength boundaries ensures consistency between instruments and experimental conditions.

Overall, the comparative analysis confirmed that slope-based variable selection achieves superior sensitivity to degradation-related changes while maintaining chemical interpretability and cross-system reproducibility. These results establish the method as a robust foundation for multivariate analysis and predictive modeling, which are presented in the following sections.

4.3 PCA Trajectories of Degradation

We conducted PCA using the variable sets obtained from each selection strategy. The slope-based configuration achieved the highest explained variance in the first principal component (PC1), capturing 97.48% of the total variance. In contrast, the user-defined and AUC-based strategies explained 96.63% and 81.47%, respectively. This indicates that the slope-based selection more effectively concentrated the chemical variability of the system into fewer dimensions. PCA projections are shown in Figure 4.3, where the slope-based configuration yielded the most compact, time-ordered trajectory. Table 4.1 resumes the accumulated variance in each case.

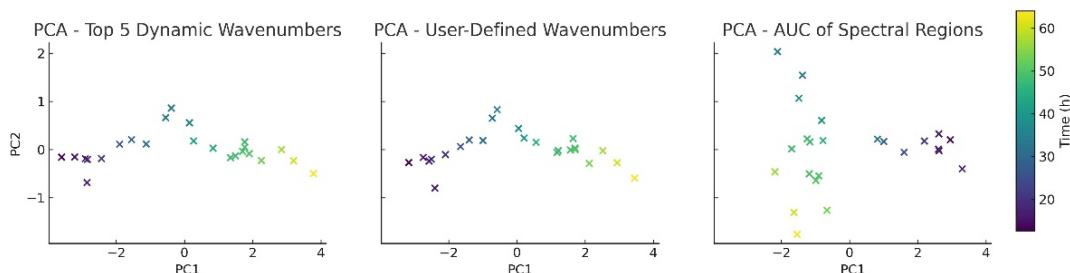


Figure 4.3: PCA projections for different variable selection methods. Colour gradient indicates sample acquisition time (hours).

Table 4.1: Explained variance by the first two principal components (PC1 and PC2)

Variable Selection Method	PC1	PC2	Total (PC1 + PC2)
Top 5 Dynamic Wavenumbers	97.48	2.22	99.69
User-Defined Wavenumbers	96.63	3.00	99.63
AUC of spectral Regions	81.47	1.679	98.26

PCA score plots revealed a strong time-ordered trajectory for slope-selected variables. Sample points were linearly distributed along PC1 according to acquisition time, illustrating the capacity of the extracted features to track the progression of degradation. The trajectories obtained from the other strategies showed greater dispersion and weaker alignment with temporal order, suggesting inferior performance in time-resolved analysis.

4.4 K-means Clustering of Degradation Stages

We applied unsupervised K-means clustering to the slope-based FTIR dataset to identify distinct stages in the thermal degradation of soybean oil. The objective was to evaluate whether the dynamic wavenumbers selected through the slope-based framework could separate early, intermediate, and advanced degradation phases based solely on spectral similarity. The analysis was performed on the reduced data obtained from the first two principal components, which together explained more than ninety-nine percent of the total spectral variance.

Following the selection criteria described in Section 3.6.2, we further evaluated the ability of each method to distinguish degradation stages using K-means clustering with $k = 3$. The slope-based configuration achieved the highest silhouette score (0.73), indicating more compact and well-separated clusters. This was followed by the user-defined (0.65) and AUC-based (0.58) methods. The corresponding inertia values confirmed that the slope-based clusters were the most cohesive.

Figure 4.4 displays the clustering outcomes, where slope-based selection provided the most distinct and compact groupings.

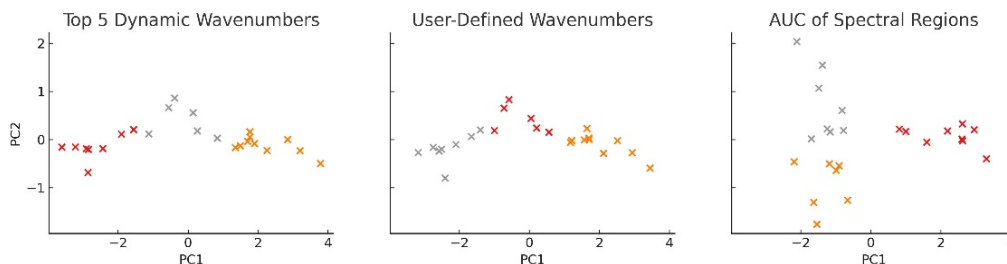


Figure 4.4: K-means clustering results based on PCA coordinates for each variable selection strategy. Colors represent assigned cluster groups.

Clustering results also correlated with sample acquisition times, allowing early, intermediate, and late stages of degradation to be distinguished without predefined labels. The spatial arrangement of the clusters reflected a natural temporal progression from unoxidized to oxidized states, consistent with the chronological order of sample collection. The transition from early to intermediate stages appeared gradual, suggesting continuous oxidative mechanisms dominated by peroxidation and conjugation. The later transition, from intermediate to advanced stages, involved sharper spectral shifts in the carbonyl and hydroxyl regions, indicating the accumulation of secondary oxidation products and potential polymerization reactions. This confirms the interpretability of the extracted features and their potential for unsupervised classification of degradation states.

4.5 Methodological Discussion

The slope-based framework provided measurable advantages over literature-defined bands and AUC integration in sensitivity, reproducibility, and interpretability. Sensitivity was evidenced by the ability of the slope profile to isolate wavenumbers with the strongest temporal response during degradation, which translated into higher correlations with chemical references and into superior multivariate performance. The slope-selected variables concentrated the largest share of relevant variance in the first principal component and produced the most time ordered PCA trajectory, indicating that they captured the underlying kinetics more efficiently than the alternatives. The clustering analysis confirmed this sensitivity at the class separation level, since the slope configuration yielded the highest silhouette score and the most cohesive clusters.

Reproducibility was strengthened by the purely data driven nature of the selection criterion. The framework relies on linear regression of absorbance as a function of time for each wavenumber and on a fixed spectral segmentation that avoids manual choices of band limits. This removes operator dependence and reduces the influence of small instrumental shifts. Because the same

numerical rule is applied to every spectrum and every segment, the selected variables are stable across replicates and transferable to other instruments configured within typical ATR tolerances. The improved stability observed in PCA loadings and in repeated clustering runs is consistent with this property.

Interpretability was preserved because each selected variable retains a direct link to a specific vibrational mode. The dynamic wavenumbers identified by the framework correspond to well established functional groups in triglycerides and oxidation products, including carbonyl stretching, C–O vibrations, unsaturation related bands, and CH rocking modes. The sign and magnitude of the slope provide an immediate indication of the direction and rate of chemical change, which facilitates mechanistic discussion and supports the connection between statistical structure and molecular processes. This contrasts with strategies that average large spectral regions or that depend on black box learning without prior feature curation.

The comparative analysis also highlighted practical implications for downstream modeling. By concentrating chemically meaningful variance into a compact set of descriptors, the slope-based framework reduces dimensionality without discarding kinetic information. This improves generalization in regression tasks, mitigates collinearity, and simplifies hyperparameter tuning. The stronger alignment between principal components, acquisition time, and the iodine value indicates that our chosen features respect the temporal ordering of the process, which is essential for robust prediction.

Two limitations should be mentioned. The linear regression used to estimate slopes assumes approximately monotonic behavior over the measurement window. Strongly nonlinear or transient responses may be underrepresented and may require piecewise or robust variants of the slope calculation. In addition, the fixed segmentation step trades spectral granularity for uniform coverage. Very narrow features can be diluted if they fall near segment boundaries, although this risk is mitigated by the ranking procedure and by the consistency of the top variables across replicates.

The results support the slope-based framework as a reproducible and interpretable strategy that enhances sensitivity to degradation related changes and improves the structure of the data for multivariate analysis. These properties justify their use as the foundation for cross domain validation and for the supervised prediction of iodine value presented in the next sections.

4.6 External Dataset Validation (Cross-Domain Robustness)

In addition to experimentally acquired spectra, we obtained supplementary datasets from previously published studies to evaluate the general applicability of our methodology. Because the original numerical spectral data were not publicly available, the FTIR spectra were reconstructed from graphical representations reported in the literature.

The extraction process was performed by converting the spectral figures into grayscale images and digitizing the spectral curves. First, each image was calibrated by defining reference points along the horizontal and vertical axes corresponding to the wavenumber and absorbance scales reported in the original figure. This calibration allowed the transformation of pixel coordinates into physically meaningful spectral values.

After calibration, the spectral curves were digitized to obtain discrete pairs of wavenumber and absorbance values. The resulting data were exported as numerical tables and interpolated onto a uniform wavenumber grid to ensure compatibility with the processing pipeline used for experimentally acquired spectra. This step guaranteed that all datasets shared the same spectral resolution and numerical format.

Finally, the reconstructed spectra were integrated into the chemometric workflow, which included preprocessing, feature extraction, and multivariate analysis. This approach allowed previously published FTIR measurements to be incorporated into the analysis while preserving the spectral trends reported in the original studies.

4.6.1 PET Crystallization Dataset

A published polyethylene terephthalate (PET) dataset was analyzed in this study to investigate the crystallization behavior of PET during a thermal cooling process using FTIR spectroscopy. The reference study by Chen et al. involved monitoring spectral changes from 270 °C to 140 °C and identified specific vibrational bands, particularly at 896, 973, 1095, 1120, 1340, and 1370 cm^{-1} , as indicators of polymer crystallization. These bands were associated with CH_2 rocking, C–O stretching, and conformational vibrations, and their intensities were interpreted semi-quantitatively to track crystallinity development.

When our slope-based framework was applied, it identified the most responsive wavenumbers during cooling as 1716, 972, 976, 1720, and 896 cm^{-1} (Figure 4.5). Notably, 896 and 972 cm^{-1} match the bands reported by Chen et al., confirming the relevance of our approach. The bands at 1716 and 1720 cm^{-1} correspond to carbonyl stretching in crystalline regions, reinforcing known markers of crystallization despite not being explicitly emphasized in the original analysis [74]. The band at 976 cm^{-1} , close to the 973 cm^{-1} marker cited by Chen et al., further supports the consistency between both approaches.

The PET dataset revealed clear and coherent spectral trends when analyzed using our slope-based framework. The selected wavenumbers showed consistent linear changes in absorbance during the cooling process. The slope magnitudes (on the order of 10^{-4}) reflect the subtle but systematic nature of crystallization. The 1716 and 1720 cm^{-1} bands exhibited positive slopes indicative of progressive crystalline domain formation, while the 972 and 896 cm^{-1} bands confirmed the

expected development of molecular order linked to CH_2 rocking modes. The absence of abrupt slope changes or inversions across the temperature range supports the interpretation of a gradual, continuous crystallization process. These findings highlight the sensitivity of the slope-based framework in capturing small but functionally significant spectral variations, offering an advantage over qualitative visual assessments.

Unlike the manual band-tracking approach used by Chen et al., our slope-based framework offers a systematic and assumption-free means to detect dynamic wavenumber regions, while objectively quantifying spectral responsiveness over time. The selected variables revealed coherent temporal trends that aligned with the expected progression of crystallization. This validates the method's ability to identify functionally relevant vibrational markers and demonstrates its usefulness in polymeric systems without requiring peak selection.

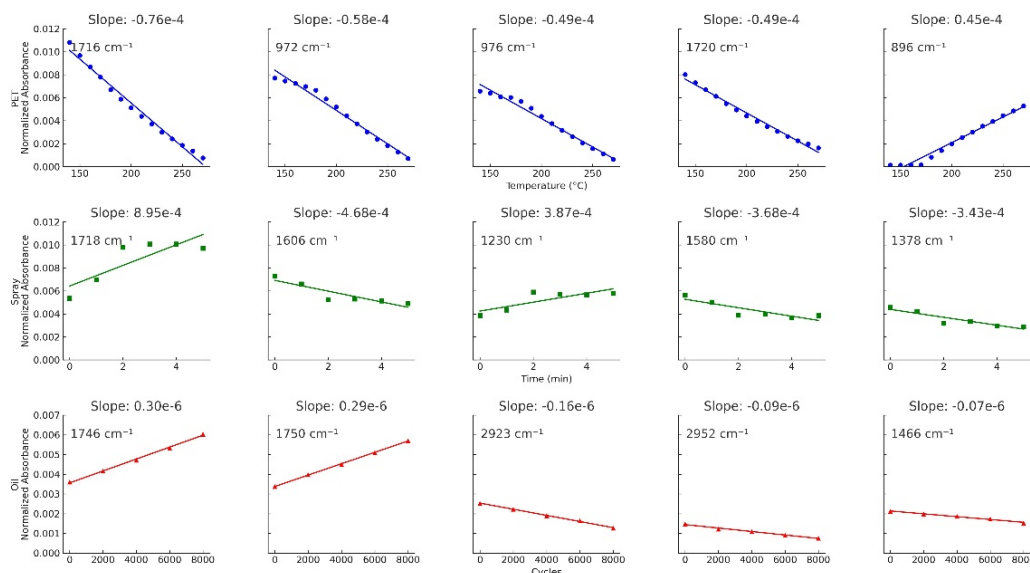


Figure 4.5: Linear regressions of selected spectral variables in the PET crystallization dataset, showing the slope magnitude and the spectral behavior as a function of temperature.

4.6.2 Spray Paint Pigment Dispersion Dataset

The spray paint dataset was obtained from the forensic study by Muehlethaler et al. (2014), which aimed to differentiate commercial paint samples based on their chemical composition using ATR-FTIR spectroscopy. The authors analyzed a wide variety of shaken paint mixtures, focusing on identifying characteristic spectral regions linked to pigments, binders, and polymeric resins [75].

In their work, spectral differentiation relied primarily on the fingerprint region ($900\text{--}1600\text{ cm}^{-1}$),

where vibrational modes from aromatic rings, C–O–C linkages, and CH bending are prominent. Specific attention was given to peaks at ~ 1358 , 1240 , and 1600 cm^{-1} , which were interpreted as indicators of pigment content and polymer matrix differences. Their approach was qualitative, combining visual inspection and known band assignments to distinguish between formulations.

Our slope-based framework identified 1718 , 1606 , 1230 , 1580 , and 1378 cm^{-1} as the most dynamic wavenumbers across cycles of paint agitation using our framework. These results align well with the spectral zones emphasized by the authors. For example, the 1606 , 1580 cm^{-1} bands correspond to aromatic C=C stretching, while 1378 cm^{-1} reflects CH bending and ring deformation—key contributors to variability in pigment-to-binder ratios. The 1230 cm^{-1} band lies close to the 1240 cm^{-1} region highlighted by the authors and is associated with C–O–C linkages in polymeric materials. The 1718 cm^{-1} signal, although outside the fingerprint region, likely relates to carbonyl stretching in polymer resins or additives.

The slope-based framework of our spray paint data showed smooth and gradual absorbance changes across key wavenumbers, with no evidence of abrupt transitions. This pattern supports the conclusion by Muehlethaler et al. that stabilization of the mixture occurs progressively, reaching reproducibility around 3 minutes of agitation. The decreasing slope magnitudes in the latter part of the time series further indicate that most of the pigment-to-binder redistribution was achieved early in the process, confirming the efficiency of mechanical homogenization within the first few minutes.

Compared to qualitative analysis in the reference study, our approach introduces a quantitative and data-driven strategy that systematically identifies the most responsive spectral regions without prior assumptions. This enhances reproducibility and offers a more sensitive method for detecting formulation dynamics, especially when small spectral changes are involved. All these results are shown in Figure 4.6.

4.6.3 Synthetic Motor Oil Degradation Dataset

The synthetic motor oil dataset was derived from the study conducted by Okubo et al., which analyzed the oxidative degradation of polyamide 66 (PA66) and lubricating oils under high-temperature conditions. The original work involved subjecting samples to thermal aging, followed by FTIR analysis to identify molecular-level alterations. The authors focused on monitoring the evolution of functional groups associated with hydrocarbon chains and oxidation products, emphasizing bands in the $2915\text{--}2865\text{ cm}^{-1}$ (C–H stretching), 1740 cm^{-1} (carbonyl stretching), and $1650\text{--}1550\text{ cm}^{-1}$ (amide) regions [76].

In their interpretation, Okubo et al. linked the formation of carbonyl and amide functionalities to the oxidation and structural breakdown of lubricant components. Their analysis was guided by

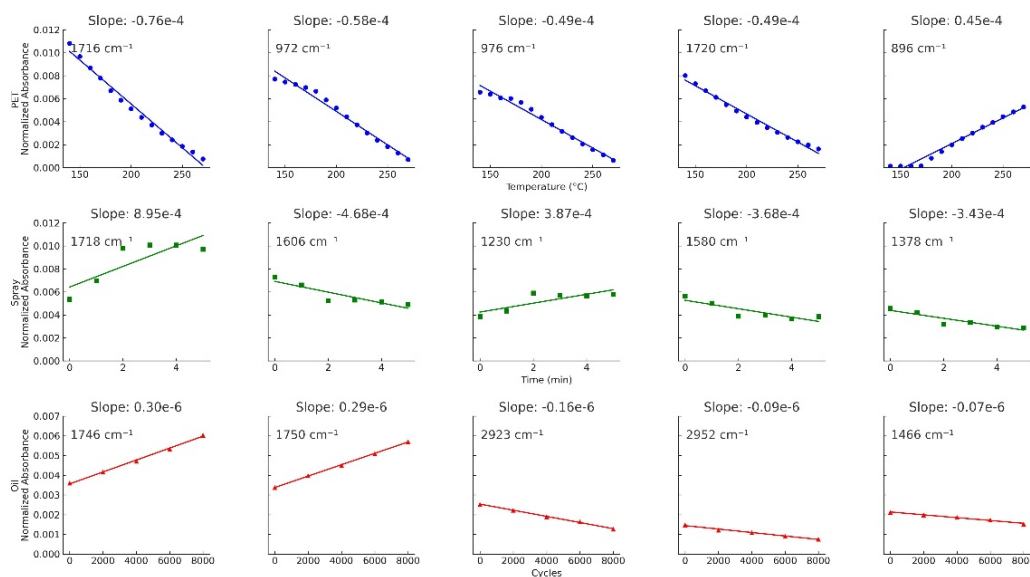


Figure 4.6: Linear regressions of selected spectral variables in the spray, showing the slope magnitude and the spectral behavior as a function of time.

known degradation markers and involved visual inspection and assignment of spectral shifts.

We processed the temporal spectrum to compute the slope at each wavenumber. The five most dynamic wavenumbers identified were 1746, 1466, 1163, 1232, and 1099 cm^{-1} . These results showed good agreement with the authors' findings: the 1746 cm^{-1} band corresponds to carbonyl groups formed during oxidation, while the 1466 cm^{-1} peak reflects CH deformation associated with hydrocarbon chain degradation (Figure 4.7). The bands at 1163, 1232, and 1099 cm^{-1} are likely related to C–O stretching and ester functionalities that are sensitive to oxidative processes, extending the spectral markers considered in the original study.

In contrast to the targeted peak tracking in the original study, our slope-based framework provided a broader, data-driven approach to identify the most responsive spectral regions. The method confirmed key degradation markers while also highlighting additional bands potentially overlooked by fixed-band approaches. This demonstrates the utility of our method in characterizing thermally degraded lubricants with minimal a priori assumptions.

4.6.4 Whey protein concentrate adulteration

This dataset corresponds to ATR–FTIR spectra of whey protein concentrate (WPC) samples adulterated with levels of wheat flour, originally investigated by Martins et al. [77]. In the reference study, wheat flour adulteration was shown to induce an overall increase in carbohydrate-related ab-

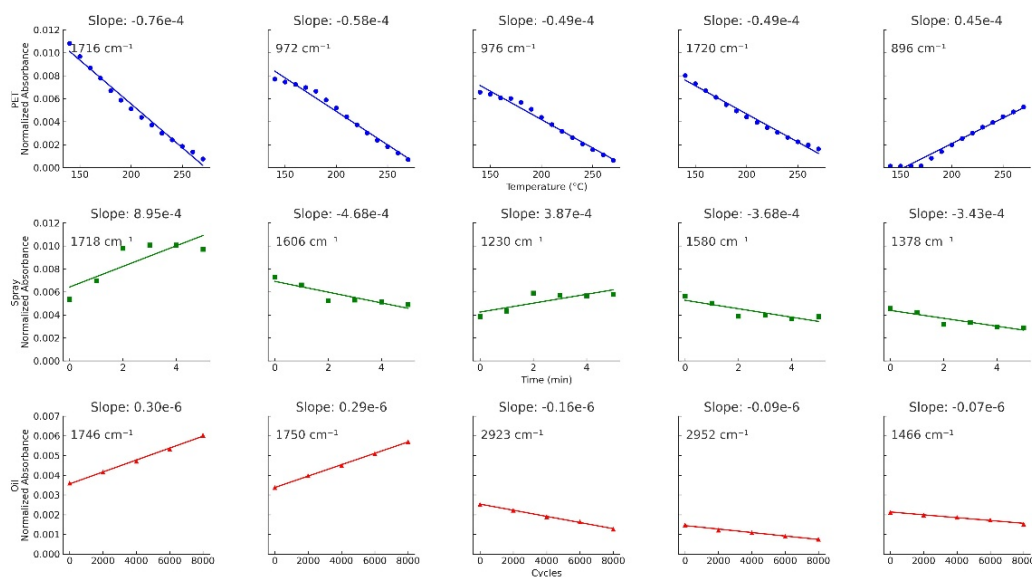


Figure 4.7: Linear regressions of selected spectral variables in the synthetic oil, showing the slope magnitude and the spectral behavior as a function of cycles.

sorbance within the fingerprint region, particularly around 1080 cm^{-1} , accompanied by a reduction in protein-associated amide bands. In our analysis, spectral variables were processed after AUC normalization, which emphasizes relative compositional changes rather than absolute intensity variations.

In agreement with these observations, the present slope-based framework highlights wavenumbers located in the same polysaccharide-dominated spectral domain emphasized in the original work. The five most responsive markers (978 , 976 , 1071 , 1025 , and 1163 cm^{-1}) automatically detected by the algorithm fall within the $1200\text{--}900\text{ cm}^{-1}$ region, which reflects polysaccharide-related vibrational changes induced by progressive wheat flour addition. Thus, while the original study focuses on representative carbohydrate and amide bands, our approach provides a complementary localization of progression-responsive features within the same chemically relevant region.

Linear regression analysis across the full adulteration range ($0\text{--}100\text{ wt\%}$) confirms that all selected markers exhibit systematic trends (Figure 4.8), validating their suitability as progression-responsive indicators of wheat flour incorporation in WPC

Our method offers a distinctive advantage by quantitatively identifying the most responsive bands based on spectral dynamics rather than expert selection. This reduces bias and captures variations induced by mechanical dispersion or formulation differences, making it suitable for forensic or quality control applications where unsupervised classification is desirable.

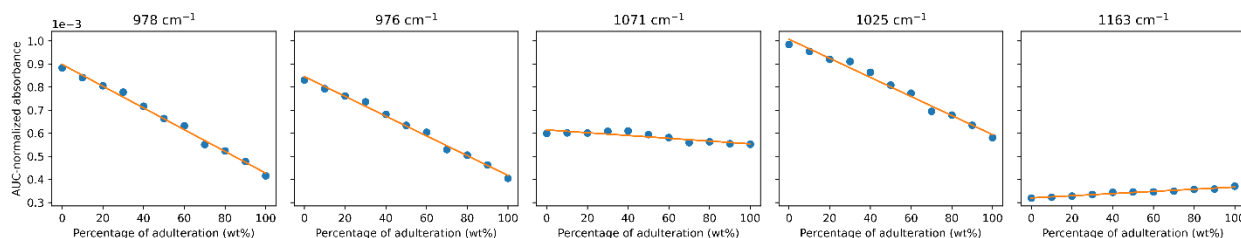


Figure 4.8: Linear regressions of selected spectral variables in the whey, showing the slope magnitude and the spectral behavior as a function of percentage of adulteration.

4.6.5 Generalizability and Limitations of the Linear Slope Approach

The external validations show that the slope based framework transfers beyond edible oils and retains diagnostic value in systems with different chemistries and kinetics. In the PET crystallization dataset, the method emphasized wavenumbers linked to chain ordering and carbonyl environment, reproducing the expected monotonic evolution of crystallinity proxies. In the synthetic motor oil experiment the selected variables tracked oxidation and additive depletion despite the presence of complex baselines and viscosity changes. In the spray paint dispersion study the framework identified regions associated with binder rearrangement and pigment environment even under heterogeneous matrices. These outcomes indicate that a simple rate of change model can recover chemically meaningful dynamics across domains and that the segment-based selection yields stable descriptors with minimal tuning.

The principal strength that enables this generalization is the reliance on a time aligned regression at each wavenumber, which requires no prior knowledge of band positions and imposes a single numerical rule across instruments and sample types. Because the criterion operates locally in the spectral axis, it remains effective when absolute intensities vary due to matrix effects or contact differences, provided that preprocessing ensures baseline consistency and variance stabilization. The uniform segmentation further supports transfer by distributing candidates across the spectrum and by limiting redundancy that can amplify instrument specific noise.

Two methodological constraints define the limits of applicability. The first is the linearity assumption in time. The model performs best when spectral responses evolve approximately monotonically within the acquisition window. Processes that exhibit induction periods, oscillatory behavior, or rapid phase transitions can produce local curvatures that a single global slope cannot represent faithfully. In such cases a piecewise formulation with change point detection, a robust Theil Sen estimator, or a local regression over sliding windows can capture nonlinear segments while preserving the original selection logic. The second constraint is the fixed width segmentation. Uniform bins guarantee coverage but can dilute very narrow features near boundaries or overweight broad plateaus. Adaptive bins informed by peak density or by local variance can alleviate this issue, as

can a refinement step that reoptimizes the final set around the provisional maxima.

Additional factors influence transfer across laboratories. Differences in spectral resolution, apodization, and ATR crystal material can shift peak shapes and penetration depth, which affects slope magnitude even when trends agree in sign. Calibration transfer methods such as standard normal variation and direct standardization, together with routine background control and humidity management, reduce these effects. Temporal sampling also matters. Sparse time grids weaken slope estimation and increase uncertainty, while irregular intervals bias least squares estimate toward densely sampled regions. Uniform timing or weighted regression restores balance in such scenarios. Finally, systems with strong outliers or intermittent artifacts require robust estimators and automated quality control to protect the selection from spurious slopes.

Despite these limits the empirical evidence from the three external datasets suggests that the framework is broadly portable. The selected variables remained interpretable, the principal components preserved temporal alignment, and the clustering structure was coherent with known stages. For predictive modeling the compact and chemically anchored descriptor set reduced dimensionality without discarding kinetics, which supports generalization in cross domain training. Future extensions can further strengthen transferability by combining slope based FTIR variables with UV-Vis indices in a multimodal design, by quantifying uncertainty through bootstrap distributions of slopes, and by applying domain adaptation or calibration transfer when instruments or matrices change. Under these considerations, the linear slope approach provides a reproducible and interpretable baseline that generalizes well across materials while offering a clear path to handle nonlinear or highly heterogeneous scenarios.

Chapter 5

PREDICTION OF IODINE VALUE USING FTIR AND MACHINE LEARNING

5.1 Importance of Iodine Value as a Reference Metric

Iodine value (IV) is a fundamental indicator of the degree of unsaturation in lipids and a direct measure of their oxidative susceptibility. It quantifies the grams of iodine absorbed by 100 g of oil, a value proportional to the concentration of carbon–carbon double bonds in fatty acid chains. Oils with high IV contain greater proportions of polyunsaturated fatty acids (PUFAs) and are thus more prone to oxidative reactions during heating and storage, whereas those with low IV are dominated by saturated components and display superior thermal stability. As thermal degradation advances, double bonds are consumed or rearranged into conjugated systems and oxidation products, producing a progressive decline in IV that reflects the overall loss of unsaturation.

Because IV integrates the contribution of all acyl chains into a single chemically interpretable quantity, it is widely recognized as a universal descriptor of oxidative stability and compositional change. Oliveira et al. (2015) [66] demonstrated that IV, together with viscosity and induction period, correlates strongly with biodiesel oxidative stability; neural-network models trained on fatty acid methyl ester composition predicted IV with low RMSE (3.49 g I₂ / 100 g), confirming its quantitative link to degradation and its utility as a quality index [78]. Likewise, Meng et al. (2017) developed ATR-FTIR + chemometric models achieving $R = 0.9885$ and $RMSEP = 2.73$ g I₂ / 100 g, underscoring that IV can be estimated accurately from mid-infrared spectra and validating its function as a benchmark for multivariate calibration across diverse edible oils [79].

Further work has emphasized the versatility of IV as a target property in predictive spectroscopy. Yan et al. (2018) showed that portable microelectromechanical systems FTIR (MEMS-FTIR), MicroNIR, and Raman devices can predict IV in blended edible oils with $RMSEP \approx 2.2 - 2.5$ g I₂ / 100 g when variable-selection algorithms such as competitive adaptive reweighted sampling (CARS) or bootstrap soft shrinkage (BOSS) are applied, demonstrating that even compact instruments can reliably infer IV and, by extension, unsaturation chemistry [80]. This confirms IV robustness as a transferable reference for both laboratory and field conditions.

The recent study by Panchbhai and Lanjewar (2025) extends these findings to food authentication: ATR-MIR + machine-learning models discriminated pure and adulterated groundnut oils with $R^2 = 0.999$ and $RMSE = 0.145$ mL (v/v), where the model performance relied on spectral features directly tied to IV differences between oleic- and palmitic-rich oils [81]. The study highlights

that IV variations arising from altered unsaturation profiles serve as the principal signal enabling detection of adulteration, oxidation, and compositional shifts.

These results illustrate that IV remains not only as a traditional chemical constant but also as a central quantitative bridge between molecular composition and spectroscopic observables. In this thesis, IV functions as the reference target for supervised prediction. It is determined independently by a validated spectrophotometric procedure and paired with time-resolved FTIR spectra processed through the slope-based framework. This integration ensures that the statistical relationships learned by the models reflect authentic chemical transformations and that predictive accuracy is assessed against an established, chemically meaningful index of oxidative stability.

Table 5.1: Comparative summary of studies predicting iodine value (IV) from spectroscopic and chemometric methods.

Study / Year	Sample Type	Technique	Model(s)	Spectral Range / Variables	Variable Selection / Preprocessing	Accuracy Metrics	Key Findings
Barradas Filho et al., 2015	Biodiesel (FAME mixtures)	Compositional (no spectra)	Artificial Neural Network (MLP)	13 fatty acid methyl ester (FAME) percentages (C8–C22)	k-fold cross-validation (k = 11)	RMSE = 3.49 g I ₂ /100 g, R ² ≈ 0.98 (test set)	IV correlated strongly with unsaturation degree; ANN outperformed linear regressions.
Meng et al., 2017	245 edible oils (soybean, peanut, rapeseed, etc.)	ATR-FTIR (mid-IR)	PLS, iPLS, fiPLS, biPLS	4000–1000 cm ⁻¹ (full spectrum split into 34 intervals)	Interval selection (8 key regions: 4000–3700, 3600–3400, 3200–3000, 2800–2700, 2300–2100, 1700–1600, 1500–1400, 1100–1000 cm ⁻¹)	RMSECV = 2.68, RMSEP = 2.73, R = 0.9885	biPLS model achieved highest accuracy; IV model was oil-type dependent but generalizable.
Yan et al., 2018	Mixed edible oils (soybean, olive, peanut, blends)	MEMS-FTIR, MicroNIR, Raman	PLS + MCUVE, CARS, BOSS, VCPA	FTIR 1100–2100 nm; NIR 900–1700 nm; Raman 175–3200 cm ⁻¹	Variable elimination and ranking (< 10% of total features)	RMSEP = 2.2185–2.5058 g I ₂ /100 g, R ² > 0.98, RPD > 9	Portable spectrometers with variable selection achieved precise on-site IV prediction.
Yang et al., 2022	Crude oil fractions	FTIR (ATR)	PCA-SVR and Autoencoder-SVR	Full FTIR (6219 wavenumbers) → dimension reduction (PCA or autoencoder)	Dimensionality reduction to avoid overfitting; data normalized	R ² = 0.97–0.99, RMSE ≈ 1.5–3.0 units	Nonlinear SVR models using reduced FTIR features successfully predicted crude oil properties.
Panchbhai & Lanjewar, 2025	Groundnut oil + Palm olein (0–50%)	ATR-MIR (5.5–10.9 μm)	Stacking Regressor / Classifier (ML ensemble)	128 MIR wavelengths after preprocessing (SG, MSC, SNV, PCA)	SG filter + MSC + SNV + PCA for noise reduction and data compression	R ² = 0.999, RMSE = 0.145 mL (v/v), Accuracy = 100%, RPD > 100	Ensemble ML approach detected adulteration through IV-related spectral differences.

The comparative analysis in Table 5.1 shows that iodine value prediction has progressed from classical wet-chemical assays to spectroscopic and machine-learning-based methodologies capable of rapid, non-destructive evaluation of oil quality. Early models, such as Oliveira et al. (2015) [66], demonstrated that IV correlates directly with the fatty-acid composition of biodiesel and, consequently, with its oxidative stability. Although their neural-network approach achieved an accuracy with $R^2 \approx 0.98$, it relied exclusively on compositional data obtained through chromatographic analysis, making it labor-intensive and not transferable to direct spectroscopic monitoring.

Subsequent research introduced infrared and near-infrared spectroscopy combined with chemometric regression. Meng et al. (2017) and Yan et al. (2018) confirmed that mid-IR, NIR, and Raman spectra contain sufficient information to estimate IV accurately, with RMSEP values around 2–3 g I₂/100 g and correlation coefficients close to 0.99. Nevertheless, their models required the entire spectral range followed by statistical feature selection methods (iPLS, CARS, BOSS), which optimize prediction but do not ensure physical interpretability of the selected variables. These algorithms segment or rank spectral intervals empirically, so the link between a chosen region and a specific chemical process, such as the oxidation of double bonds, remains indirect.

Later works expanded the methodological complexity. Yang et al. (2022) used PCA and autoencoders coupled to support-vector regression to predict crude-oil properties from thousands of FTIR variables, and Panchbhai & Lanjewar (2025) applied stacking regressors and classifiers to detect adulteration in groundnut oil using ATR-MIR spectra. Both achieved near-perfect correlations ($R^2 > 0.99$) but relied on extensive preprocessing (SG, MSC, SNV) and high-dimensional latent variables, which hinder chemical interpretability and generalization beyond the calibration domain. In addition, these approaches analyze static spectra, overlooking the temporal evolution of oxidation and unsaturation loss that defines IV degradation kinetics.

Our methodology addresses these limitations by combining time-resolved FTIR spectroscopy with a slope-based wavenumber selection strategy. Instead of statistically partitioning the spectrum, the method identifies wavenumbers exhibiting the greatest temporal variation in absorbance intensity, directly linked to the consumption of unsaturated bonds, formation of carbonyl species, and other oxidation-driven transformations. These chemically grounded variables are subsequently used as predictors in supervised regression models (Linear, PLSR, Random Forest, SVR) to estimate the iodine value.

Our design provides four advantages over previous approaches.

1. First, each selected wavenumber corresponds to a specific vibrational mode associated with degradation chemistry, ensuring physical interpretability of model coefficients.

2. Second, only a few informative features are required, reducing overfitting risk and improving model stability, which we refer to as dimensional efficiency.
3. Third, our slope-based criterion captures the dynamics of oxidation, enabling regression against IV as a time-dependent degradation metric rather than a static indicator.
4. Fourth, because selection is governed by spectral behavior rather than oil type, our framework generalizes across vegetable oils, industrial fats, or synthetic lipid systems.

While existing studies validate the feasibility of spectroscopic IV prediction, they remain either empirically driven or limited to static datasets. Our slope-based framework integrates chemometric rigor, temporal information, and chemical interpretability within a unified analytical strategy, offering a more mechanistic and generalizable pathway for quantifying IV from FTIR data.

5.2 Integration of Slope-Selected Wavenumbers with IV Data

Each experimental sample produced a paired record consisting of one IV measurement and one FTIR spectrum. Iodine value was obtained with the Soares spectrophotometric procedure and recorded with the same sample identifier and timestamp used in optical acquisition. FTIR spectra were preprocessed through baseline correction, Savitzky Golay smoothing, area normalization, and first derivative transformation to improve resolution and stabilize baselines. Time alignment was verified by matching identifiers and by checking that the acquisition order of spectra and IV records were identical. Records with incomplete metadata or inconsistent timing were excluded to prevent label leakage and misassignment.

For feature construction, the absorbance at every wavenumber was regressed against acquisition time using ordinary least squares. The absolute value of the slope was taken as a quantitative index of spectral responsiveness. The spectrum was divided into consecutive segments of one hundred reciprocal centimeters to ensure uniform coverage across the infrared domain. Within each segment the wavenumber with the largest slope magnitude was retained as the representative candidate. From the candidate set, the five variables with the highest slope magnitude were selected as the most dynamic wavenumbers of degradation. These variables corresponded consistently to regions associated with carbonyl stretching, C-O vibrations, unsaturation related modes, and long chain ordering.

Predictor values used for modeling were the normalized absorbances of the selected wavenumbers at each time point rather than the slope magnitudes themselves. Using absorbance preserves the sign and curvature of the temporal evolution and allows direct chemical interpretation of model coefficients and loadings, while the slope serves only as an objective selection criterion. Before modeling, predictors were centered and scaled within each training fold to unit variance to avoid

scale driven bias. The response vector contained the paired IV measured on the same samples. Data matrices were inspected for collinearity and for undue leverage points using variance inflation factors and Cook’s distance statistics, and no record exceeded conservative thresholds.

The integrated dataset therefore consists of a compact, chemically interpretable descriptor set aligned one to one with IV. This construction links dynamic spectral markers to a validated chemical index while controlling timing, scaling, and data integrity. It provides a robust foundation for supervised learning, enabling fair comparison among linear and nonlinear models and ensuring that performance reflects the ability to recover a recognized measure of unsaturation from FTIR information.

The IV measurements used as targets followed the Soares spectrophotometric procedure with triplicate repeats and documented uncertainty as described in Section 3.2, which provides the traceable ground truth for all subsequent modeling.

5.3 Results and Model Performance

The data set comprised 108 paired observations drawn from two sequences of 54 samples each. The split used for evaluation contained 64 rows for training, 22 rows for validation, and 22 rows for testing. All models were fitted with the same preprocessing and assessed with identical metrics to allow a fair comparison.

Parity plots on the held-out test set show that ensemble learners provided the closest agreement with laboratory IV. The perfect training score of Extra Trees (Train $R^2 = 1.000$, Train RMSE = 0.000) reflects the tendency of this algorithm to memorize training samples when no depth limit is imposed; however, the strong test performance ($R^2 = 0.993$) confirms that this did not translate into overfitting on unseen data, likely because the compact five-variable descriptor set inherently constrains the hypothesis space. Extra Trees achieved a test coefficient of determination of 0.993 with a root mean square error of 3.127 IV units. XGBoost obtained a test coefficient of determination of 0.985 with a root mean square error of 4.532. Random Forest reached a test coefficient of determination of 0.959 with a root mean square error of 7.541. Gradient Boosting produced a test coefficient of determination of 0.948 with a root mean square error of 8.452. Ridge and linear baselines performed noticeably worse on the test split, with ridge yielding a coefficient of determination of 0.911 and a root mean square error of 11.077, and linear regression dropping to 0.684 with a root mean square error of 20.838.

Validation behavior differed by family. Linear and ridge models displayed high validation coefficients near 0.94 but degraded on the test set, indicating that a purely linear mapping does not capture the curvature present in the relation between spectral markers and IV. Tree based ensembles showed more modest validation scores but generalized strongly to the test set, consistent with their ability

to model interactions among the selected wavenumbers. The bar chart that compares validation and test metrics summarizes this pattern and confirms the stability of the best ensembles.

Calibration was inspected through residuals against fitted values. Extra Trees and XGBoost produced residuals centered near zero across the full IV range without visible heteroscedasticity. Gradient Boosting showed slightly wider dispersion at low IV but no systematic bias. Linear and ridge models exhibited structured residuals that reflect underfitting of nonlinear effects. The dispersion of cross validated errors remained within the analytical uncertainty reported for the spectrophotometric reference 3.2 Reference Chemical Assays for the top ensembles, which sets a practical floor for attainable accuracy.

Interpretability remained consistent with the chemistry of thermal oxidation. In linear and ridge models the signs of standardized coefficients were negative for carbonyl and C-O regions and positive for unsaturation and chain order markers. Variable importance from Random Forest and Extra Trees and permutation importance from XGBoost converged on the same subset of influential wavenumbers, with carbonyl and C-O contributions dominating the prediction and unsaturation markers supporting positive adjustments. This convergence across model families supports internal validity and explains the strong calibration observed in the parity plots in Figure 5.1.

Model performance is summarized in Table 5.2. The best results were obtained with Extra Trees and XGBoost, followed by Random Forest and Gradient Boosting, while linear baselines underperformed on the held-out test set.

Table 5.2: Cross validated and test performance for iodine value prediction from slope selected FTIR variables.

Model	Test R ²	Test RMSE	Test MAE	Train R ²	Train RMSE	Train MAE	Val R ²	Val RMSE	Val MAE	CV R ² Mean	CV R ² Std
Extra Trees	0.993	3.127	1.951	1.000	0.000	0.000	0.900	8.574	3.375	0.971	0.020
XGBoost	0.985	4.532	2.779	0.999	0.665	0.395	0.792	12.376	4.674	0.971	0.020
Random Forest	0.959	7.541	4.142	0.997	1.438	0.974	0.528	18.667	6.430	0.972	0.009
Gradient Boosting	0.948	8.452	4.350	1.000	0.154	0.126	0.568	17.845	6.034	0.977	0.011
Ridge	0.911	11.077	8.121	0.870	10.107	7.623	0.941	6.624	5.355	0.825	0.029
Linear Re- gression	0.684	20.838	11.034	0.901	8.803	7.139	0.934	7.001	5.318	0.794	0.097

Overall, Extra Trees and XGBoost provided the most accurate and stable predictions under the defined split. Their test errors are small in absolute terms, and their residuals show no systematic trends. The agreement between importance rankings and known chemical behavior indicates

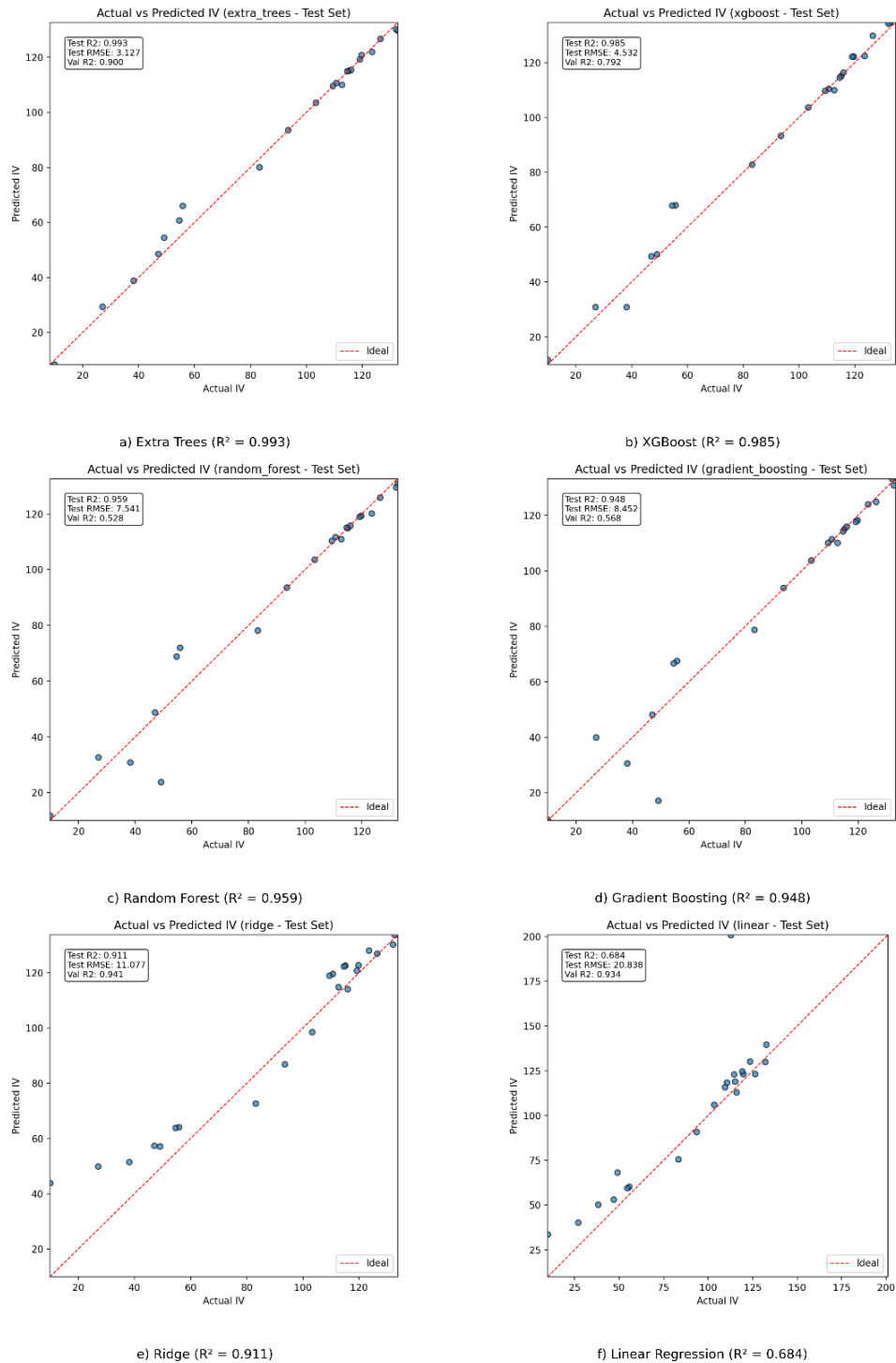


Figure 5.1: Comparison between measured and predicted iodine value (IV) for different regression models on the test set. Panels are ordered by decreasing test R^2 : (a) Extra Trees, (b) XGBoost, (c) Random Forest, (d) Gradient Boosting, (e) Ridge regression, and (f) Linear regression. The dashed red line represents the ideal 1:1 relationship between predicted and actual IV values.

that the learned relations are chemically plausible. These findings justify selecting one of these ensembles as the reference predictor for the discussion of predictive potential in Section 5.5.

5.4 Discussion of Predictive Potential

The results demonstrate that IV can be recovered from a compact set of slope-selected FTIR descriptors with accuracy and calibration that are competitive with reference laboratory method. The best ensembles achieved test errors of a few IV units and coefficients of determination above 0.98, as reported in Table 5.1 and illustrated in the parity plots. Residuals remained centered across the full range of IV without visible heteroscedasticity, which indicates that the learned relationships are stable within the experimental domain. Convergence of variable importance across algorithms toward carbonyl and C-O regions, together with positive contributions from unsaturation markers, supports the chemical plausibility of the predictors and explains the strong calibration observed.

When contrasted with traditional chemical assays, the spectroscopic approach offers practical advantages. Measurement is nondestructive, requires minimal preparation, and can be executed in minutes using the same FTIR acquisition already performed for process monitoring. The elimination of titrants and the reduction of consumables align with principles of green analytical chemistry and reduce per sample cost. At the same time, the reference assay remains essential as the traceable anchor for calibration and periodic verification. The comparison between predicted and measured IV should therefore be interpreted relative to the analytical uncertainty of the spectrophotometric procedure described in Section 3.2, which sets a practical floor for RMSE and MAE. Our top models approached that floor, which suggests that further gains may be limited more by reference noise than by modeling capacity.

The predictive system has clear industrial implications. Integrated into a quality control workflow, FTIR based estimation of iodine value can provide near real time feedback on the oxidative state of oils used in production. This capability enables earlier detection of degradation, supports dynamic adjustments of heating protocols, and informs replacement decisions before quality thresholds are crossed. Because the descriptor set is small and chemically interpretable, model maintenance is straightforward. Routine checks can compare current parity plots against those in Figure 5.1 and trigger recalibration if drift is detected. A practical strategy is to rely on FTIR prediction for high frequency screening while confirming a subset of samples with the chemical reference at scheduled intervals to maintain traceability.

Two considerations delimit the operating envelope of the models. First, validity is bounded by the domain of training. Changes in oil variety, additive package, temperature regime, or instrumental configuration can alter spectral responses and require verification before deployment. Calibration transfer methods and a short transfer set measured with the reference assay can restore performance

when instruments or matrices change. Second, the slope based selection assumes approximately monotonic evolution during the acquisition window. Processes with induction periods or pronounced non linearities may benefit from piecewise slope estimation or local regression, which can be incorporated without altering the interpretability of the framework.

Within these boundaries, the evidence supports adoption of the spectroscopic predictor as a practical complement to the chemical assay. It reduces analysis time and reagent use, preserves sample integrity, and provides interpretable diagnostics through variable importance and loadings. The combination of rapid estimation with periodic chemical confirmation achieves a balance between efficiency and traceability that is well suited to industrial monitoring and to research workflows that require dense temporal sampling of oxidative dynamics.

Chapter 6

CONCLUSIONS AND FUTURE PERSPECTIVES

6.1 Summary of Findings

In this thesis, we addressed the challenge of reliably monitoring degradation and progression systems in edible oils using spectroscopic data, with particular emphasis on overcoming the subjectivity, limited reproducibility, and poor transferability associated with conventional FTIR variable selection strategies. To resolve this limitation, we developed and validated a slope-based framework for FTIR data processing as a quantitative and data-driven approach for identifying chemically meaningful spectral variables in progression system datasets.

Our framework quantifies the rate of absorbance change at each wavenumber through linear regression across time and ranks variables according to slope magnitude, enabling the objective identification of dynamic wavenumber regions without relying on predefined band assignments. This strategy directly addresses the problem of empirical feature selection in high-dimensional and highly collinear FTIR data, providing a reproducible mechanism for dimensionality reduction while preserving chemical interpretability.

When applied to thermally degraded soybean oil, the framework isolated a compact set of dynamic wavenumbers, including 1731, 988, 709, 699, and 1112 cm^{-1} , which exhibited monotonic temporal behavior and corresponded to well-established vibrational modes associated with carbonyl formation, C–H deformation, and C–O stretching. Compared with traditional selection approaches based on fixed bands or integrated spectral regions, the slope-based framework achieved superior performance in multivariate analysis, explaining 97.48% of the variance in the first principal component and producing coherent clustering of degradation stages with a silhouette score of 0.73. These results demonstrate that chemically meaningful information can be condensed into a small and interpretable descriptor set while preserving the temporal structure of the degradation process.

We further confirmed the generalizability of our framework through validation on three independent progression-based datasets involving PET crystallization, synthetic motor oil oxidation, and spray paint dispersion. In all cases, slope-selected wavenumbers were consistent with literature-reported spectral markers, including bands at 896 and 972 cm^{-1} for PET, 1746 and 1466 cm^{-1} for lubricant oxidation, and 1606 and 1230 cm^{-1} for paint formulations. This cross-domain agreement demonstrates that the method captures process-dependent spectral dynamics rather than system-specific artifacts, supporting its transferability beyond edible oil matrices.

Beyond qualitative and exploratory analysis, the slope-based framework enabled accurate prediction of iodine value from FTIR data using a reduced and chemically interpretable set of descriptors. The selected variables reflected key degradation phenomena, including unsaturation loss and carbonyl growth, and served as effective inputs for machine learning regression models. Across models trained under an identical preprocessing and validation pipeline, ensemble learners exhibited the strongest predictive performance. Extra Trees achieved a test coefficient of determination of 0.993 with a root mean square error of 3.127 iodine value units, while XGBoost achieved a test coefficient of determination of 0.985 with a root mean square error of 4.532. These results indicate that nonlinear relationships between dynamic spectral features and iodine value are relevant within the experimental domain and can be reliably captured when variable selection is driven by spectral responsiveness.

Residual analyses for the best-performing models showed centered error distributions without evident heteroscedasticity, and predictive accuracy approached the analytical uncertainty of the spectrophotometric reference method used as ground truth. These findings directly address the limitation identified in the problem statement regarding the integration of spectroscopy with machine learning for chemical index prediction, demonstrating that accurate and robust iodine value estimation is achievable using non-destructive FTIR measurements when feature selection is objective and progression aware.

The results confirm that the proposed slope-based framework resolves the core limitations associated with empirical FTIR variable selection by providing a quantitative, reproducible, and assumption-free methodology for spectral feature extraction. The framework improves dimensionality reduction, enhances temporal interpretability, and supports cross-system application, data-driven spectroscopic analysis pipelines applicable to chemical monitoring, quality control, and degradation studies across a wide range of materials.

6.2 Original Contributions of the Thesis

Our work makes four distinct contributions to the field of spectroscopic data analysis. The first is the slope-based framework itself: a quantitative, data-driven method for selecting dynamic FTIR variables by ranking wavenumbers according to their rate of absorbance change, providing a reproducible and assumption-free alternative to empirical band selection. The second contribution is the demonstration that this framework generalizes across chemically diverse progression systems, including vegetable oil degradation, polymer crystallization, lubricant oxidation, paint dispersion, and food adulteration, confirming its transferability beyond a single application domain. The third contribution is the integration of slope-selected variables with supervised machine learning for iodine value prediction, showing that a compact five-variable descriptor set can match the accu-

racy of full-spectrum models while preserving chemical interpretability. The fourth contribution is a fully automated Python-based workflow that implements preprocessing, slope-based feature selection, multivariate analysis, and regression in a reproducible and modular pipeline, making the methodology directly applicable to new datasets without manual parameter tuning. We validated this strategy across diverse materials and established a quantitative foundation for feature extraction by ranking wavenumbers according to their responsiveness, providing an interpretable and reproducible alternative to traditional selection based on empirical or literature-defined bands. The results demonstrate that slope-selected wavenumbers improve dimensionality reduction and clustering performance, enabling clearer temporal and chemical differentiation in degradation processes. This is for non-destructive, data-driven monitoring of transformation and degradation processes using FTIR spectroscopy.

6.3 Limitations of the Present Work

Despite its robustness and versatility, the slope-based framework presents certain limitations that should be acknowledged. Because the method relies on linear regression to quantify spectral change, it may underrepresent non-linear or transient behaviors that occur in complex reactions, such as oscillatory or multi-step degradation pathways. The segmentation by fixed spectral intervals, while effective for ensuring coverage, can also overlook subtle variations within narrow overlapping bands. The selection of the most dynamic wavenumbers also depends on the temporal resolution of the dataset, which means that irregular sampling intervals or limited data points could reduce sensitivity. Finally, the current implementation focuses on univariate slope computation and does not yet incorporate multi-dimensional interactions between correlated spectral regions. These limitations suggest potential directions for refinement through adaptive segmentation, non-linear modeling, or multi-slope correlation analyses in future work.

6.4 Future Work

Future research should focus on extending the slope-based framework toward more adaptive and generalized forms of spectral trend analysis. One promising direction is the incorporation of non-linear regression or machine learning models capable of capturing complex temporal behaviors and multi-phase transformations that cannot be fully described by simple linear slopes. Another area of interest is the integration of dynamic segmentation, where spectral regions are adaptively defined based on local variance rather than fixed intervals, improving sensitivity to subtle spectral shifts. Expanding the framework to include multi-slope correlation analysis and higher-order derivatives could further enhance its interpretive power by allowing the detection of coupled vibrational modes and overlapping functional group dynamics.

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